

Intrusion detection systems in IoT: A detailed review of threat categories, detection strategies, and future technologies

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ABSTRACT

The rapid growth of the Internet of Things (IoT) has transformed numerous sectors by enabling enhanced connectivity and automation among devices in industrial settings. However, this expansion has brought forward notable security concerns, as Internet enabled and connected devices has become increasingly vulnerable to a variety of cyberattacks. This has elevated the importance of Internet of Things security, necessitating robust defense mechanisms. In this paper, we thoroughly examine Intrusion Detection Systems (IDS) within the context of IoT networks, focusing on the different types of attacks and the corresponding detection methods designed to counteract them. Specifically, we classify IoT-specific threats into categories such as network based, device-level, data-centric, and insider attacks, providing insights into their mechanisms, impacts, and real-world occurrences. To address these threats, various IDS approaches are discussed, including signature based IDS, anomaly based IDS, specification based IDS, and hybrid IDS techniques. We further explore the application of Machine Learning in enhancing IDS capabilities for Internet of Things security. Each method's strengths and limitations are evaluated in terms of accuracy, adaptability, computational efficiency, and scalability. By exploring emerging trends, ongoing challenges, and potential future directions in IDS research for IoT, this study underscores the urgent need for adaptive, scalable, and effective IDS frameworks to protect IoT ecosystems against evolving cyber threats. In addition, this survey provides a critical assessment of the current research landscape, highlighting the fundamental challenges that remain unresolved and outlining future research directions derived both from the existing literature and our own domain-specific analysis.

1. Introduction

The IoT has transformed various industries by seamlessly connecting physical devices to the internet, allowing for extensive data collection, real-time analysis, and automation [1,2]. As a foundational component in areas such as smart cities, healthcare, industrial automation, and smart homes, IoT technology has significantly enhanced operational efficiency and innovation [3–5]. However, the proliferation of IoT devices has also led to an increase in security vulnerabilities, turning IoT systems into attractive targets for cyber-attacks [6]. A major challenge in securing IoT networks stems from the wide array of technologies involved, each with unique security requirements and potential vulnerabilities [7].

Low-power wireless communication technologies such as IEEE 802.15.4, ZigBee, BLE, and 6LoWPAN are extensively used due to their energy efficiency, especially in resource-constrained environments like sensor networks and smart homes [8,9]. These protocols support the

deployment of large-scale IoT networks, but their limited computational resources often hinder robust security implementations, making them susceptible to attacks such as eavesdropping, replay attacks, and DoS [10]. Thus, the balance between low energy consumption and security remains a critical concern in IoT. On the other hand, long-range communication technologies such as LoRaWAN, Sigfox, NB-IoT, and others like LTE-M and 5G IoT further extend the connectivity of IoT networks, enabling communication over vast distances [11]. These technologies are particularly valuable in applications such as smart agriculture, environmental monitoring, and infrastructure management, where devices are often deployed in remote or hard-to-access locations [12–15]. However, despite their advantages in covering large areas with low power consumption, these protocols often rely on lightweight encryption and authentication mechanisms to optimize energy efficiency [16]. This trade-off between energy efficiency and security leaves IoT systems vulnerable to various attack vectors, including message spoofing, replay attacks, and network hijacking [17–20].

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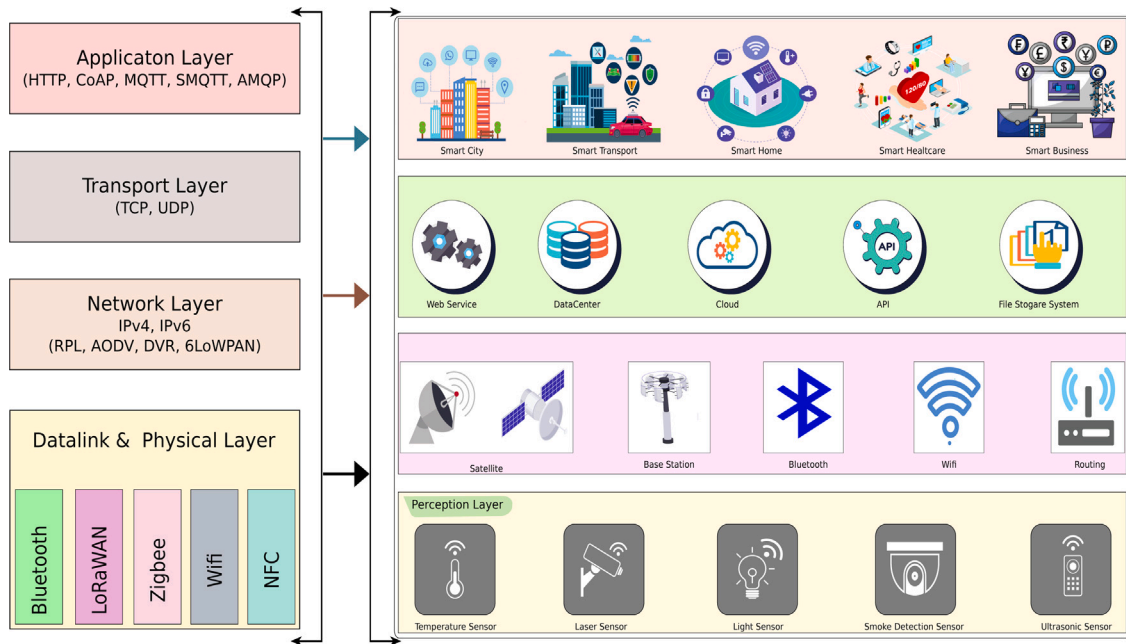


Fig. 1. Layers in IoT system.

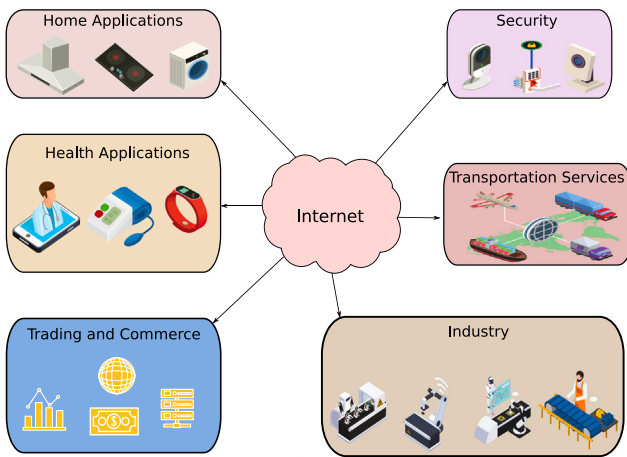


Fig. 2. Internet of things application areas [24].

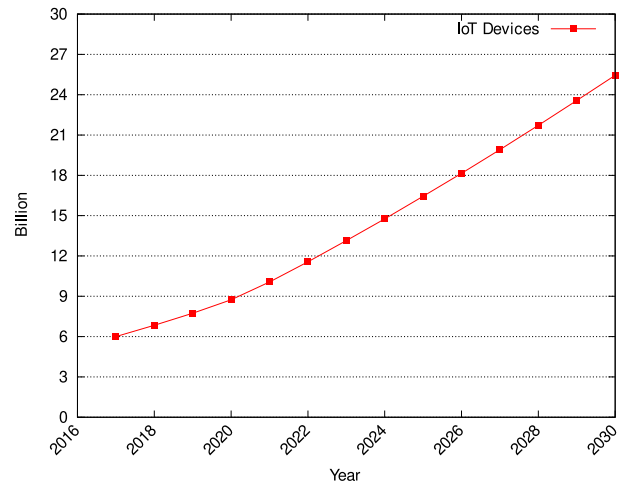


Fig. 3. Internet of things device installation trends [27,28].

The diversity and technical complexity of IoT technologies entails the use of several acronyms. Hence, a list of acronyms used throughout this study is provided in Table 1 for reference. Furthermore, to better understand the root of the security challenges faced by an IoT system, it is helpful to examine its communication architecture which is commonly structured into five distinct layers, as depicted in Fig. 1. These layers are named as physical, data link, network, transport, and application according to the TCP/IP protocol stack [21–23]. Each layer plays a critical role in enabling IoT functionalities. The data link and physical layers ensure reliable data transmission over various communication technologies. The network and transport layers manage data routing, addressing, and end-to-end communication. Finally, the application layer delivers services tailored to specific IoT use cases such as smart cities, healthcare, and industrial automation. A comprehensive understanding of these layers is essential for developing secure, scalable, and efficient IoT systems.

As illustrated in Fig. 2, IoT is widely deployed across various domains, including healthcare, industrial automation, and smart cities, each with distinct security requirements. In healthcare, the protection

of IoT devices is directly linked to patient safety, while in industrial settings, security breaches can lead to operational shutdowns, highlighting the importance of tailored security solutions for different IoT applications. Moreover, Fig. 3 emphasizes the exponential growth in the number of IoT devices deployed over the years [25]. This rapid expansion presents significant challenges in securing these networks, as the increase in connected devices creates a broader attack surface for cyber-criminals [26]. With the IoT ecosystem continually expanding, managing and securing these networks is becoming increasingly complex, necessitating advanced IDS solutions that can scale and adapt to this evolving threat landscape.

In addition, as seen in Fig. 4, the number of attacks on IoT devices is projected to rise sharply by 2030. This trend underscores the critical need for IDS solutions capable of effectively safeguarding IoT infrastructures from diverse and sophisticated cyber threats. Through this survey, we aim to provide a comprehensive examination of the latest IDS methodologies, assessing their effectiveness in mitigating key security risks and offering insights into the future direction of IDS development for IoT networks.

Table 1
List of acronyms.

Acronym	Description
6LoWPAN	IPv6 over Low-Power Wireless Personal Area Networks
AIDS	Anomaly based Intrusion Detection System
AODV	Ad hoc On-Demand Distance Vector Routing
AMQP	Advanced Message Queuing Protocol (for reliable messaging)
API	Application Programming Interface
BFT	Byzantine Fault Tolerance
BLE	Bluetooth Low Energy
CNNs	Convolutional Neural Networks
CoAP	Constrained Application Protocol
COSE	CBOR Object Signing and Encryption
CPU	Central Processing Unit
CSMA/CA	Carrier Sense Multiple Access With Collision Avoidance
DAE	Deep Autoencoder
DANN	Domain-Adversarial Neural Network
DDS	Data Distribution Service
DDoS	Distributed Denial-of-Service
DL	Deep Learning
DLT	Distributed Ledger Technology
DoS	Denial-of-Service
DVR	Distance Vector Routing
EFSM	Extended Finite State Machine
FPR	False Positive Rate
FL	Federated Learning
GANs	Generative Adversarial Networks
GWO	Grey Wolf Optimization
HGB	Histogram Gradient Boost
IDS	Intrusion Detection System
IIoT	Industrial Internet of Things
IoT	Internet of Things
IP	Internet Protocol
KNN	K-Nearest Neighbors
LOF	Local Outlier Factor
LoRaWAN	Long Range Wide Area Network
LTE-M	Long Term Evolution for Machines
LSTM	Long Short-Term Memory
MANETs	Mobile Ad Hoc Networks
MC-MitM	Multi-Channel Man-in-the-Middle
MFA	Multi-Factor Authentication
ML	Machine Learning
MSDL	Multi-Stage Deep Learning
MQTT	Message Queuing Telemetry Transport
NB-IoT	Narrowband Internet of Things
OLSR	Optimized Link State Routing Protocol
PQC	Post-Quantum Cryptography
RBAC	Role Based Access Control
RF	Random Forest
RNNs	Recurrent Neural Networks
RPL	Routing Protocol for Low-Power and Lossy Networks
R2L	Remote to Local
SMQTT	Secure Message Queuing Telemetry Transport
SVM	Support Vector Machines
TCP	Transmission Control Protocol
U2R	User to Root
VBFT	Verifiable Byzantine Fault Tolerance
WSN	Wireless Sensor Network
XMPP	Extensible Messaging and Presence Protocol
XSS	Cross-Site Scripting

1.1. Motivation

The main motivation of this survey is to provide a comprehensive guide to academicians, engineers and researchers who will conduct research in this field by analyzing the current IDS trends in the field of Internet of Things in detail. Our study examines the integration of IDS systems into the IoT ecosystem and emphasizes the importance of these systems. Firstly, we list the current security vulnerabilities in IoT landscape and investigate proposals that aim to deal with such security threats. This study primarily focuses on security vulnerabilities and attack vectors that are especially evident in low-power and resource-constrained devices. Therefore, we focus on how IDS systems can be adapted to the IoT environments and ways to increase the effectiveness

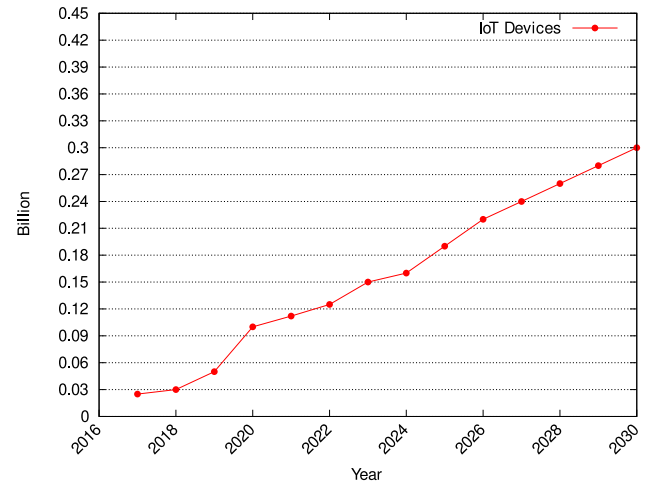


Fig. 4. IoT device attack trends [29].

of these systems. In addition, IDS methods and algorithms specifically developed for IoT are reviewed in terms of their core techniques, datasets used, evaluation metrics, and the key limitations or insights they offer. In this context, this study focuses on the following topics: current IoT security trends and threats, basic principles of IDS systems and solutions optimized for IoT. We aim to provide a broad perspective to the readers by addressing the best practices and future research directions in this field.

1.2. Contributions

This survey makes several significant contributions to the field of IoT security by addressing the integration and optimization of IDS within the IoT ecosystem. The key contributions of this study are as follows:

- **Comprehensive Analysis of Current IDS Trends:** This study provides an in-depth analysis of the latest trends in IDS specifically tailored for IoT environments. By highlighting the most recent advancements and methodologies, it serves as a valuable resource for researchers aiming to stay abreast of the current state of the field. It also briefly addresses emerging concerns such as post-quantum cryptographic risks, which remain largely unaddressed in prior IDS surveys.
- **Identification of Security Threats and Attack:** We thoroughly examine the security threats and attacks that are particularly prevalent in low-power and resource-constrained IoT devices. This detailed examination helps in understanding the unique challenges faced in securing IoT networks and provides a foundation for developing robust IDS solutions.
- **Evaluation of IDS Approaches:** The study evaluates various IDS approaches and algorithms specifically designed for IoT networks. By comparing the advantages and disadvantages of existing solutions, it identifies the most effective approaches and highlights areas that require further research and improvement.
- **Exploration of Blockchain Enabled IDS:** This work uniquely addresses the integration of blockchain into IDS design. It discusses smart contract-driven automation, decentralized anomaly logging, and federated-learning synergies, along with deployment limitations in IoT settings.
- **Guidance on Best Practices and Future Research Directions:** The survey not only addresses the current trends and solutions but also provides guidance on best practices for implementing IDS in IoT networks. Additionally, it outlines future research directions, helping to shape the development of next-generation IDS systems that are more resilient and efficient.

- **A Broad Perspective on IoT Security:** Combining analysis of current trends, evaluation of methods and practical guidance, this survey provides a broad and comprehensive perspective on IoT security. It aims to help academics, engineers and researchers develop effective security measures for IoT networks.
- **Differentiation from Existing Surveys:** In contrast to generic IDS surveys, this work focuses on constrained wireless environments (e.g., WSNs, 6LoWPAN, IIoT), yet also encompasses application layer and cloud-integrated IDS use cases, ensuring both depth and breadth of analysis.
- **Integration of New Datasets, Comparative Tables, and a Case Study:** To enhance practical relevance, this work integrates updated benchmark datasets (e.g., IoTR-DS, TON_IoT, BoT-IoT, ROUT-4-2023), comparative tables highlighting evaluation metrics and IDS architectures, and a case study that demonstrates the detection of RPL based routing attacks using eight machine learning algorithms on the ROUT-4-2023 dataset. This provides an empirical foundation supporting the survey's analytical claims.

1.3. Paper organization

The remainder of this paper is structured as follows: Section 2 presents an overview of IoT architecture and associated security issues. Section 3 details various IDS methodologies, emphasizing their applicability in IoT environments, their integration with machine learning techniques, and reviews relevant literature by comparing different IDS methods and summarizing key findings. Section 4 outlines the major shortcomings in current IoT-specific IDS research, emphasizing missing capabilities across individual protocol layers, lack of cross-layer integration, and limited applicability under real-world constraints. Section 5 highlights the key challenges faced by IDS in IoT networks and outlines potential future research directions. Finally, Section 6 concludes with a summary of findings and recommendations for future research.

2. Security challenges in IoT architecture: Threats and countermeasures

The security landscape of IoT networks is shaped not only by their heterogeneous and resource constrained architecture, but also by the layered interplay of legacy protocols, untrusted environments, and evolving threat surfaces. While Table 2 provides an overview of architectural and operational-level vulnerabilities such as energy constraints, unprotected deployment contexts, and software update limitations a deeper understanding requires analyzing how specific attack vectors exploit protocol-level weaknesses across the IoT stack. To address this, Table 5 offers a protocol-aware, layer-by-layer taxonomy that systematically maps attack types to affected technologies, known vulnerabilities, countermeasure mechanisms, and their deployment constraints.

2.1. Common attack types in IoT networks

In this section, we present a detailed taxonomy of various types of attacks in IoT systems. Each attack type presents unique challenges and requires tailored countermeasures to ensure the security of IoT networks [57]. By categorizing attacks based on the different layers of an IoT system, a clear understanding is provided of how specific vulnerabilities are exploited across the IoT architecture.

2.1.1. Physical layer attacks

Physical layer attacks exploit vulnerabilities in the *wireless* communication medium of IoT networks [58]. Unlike higher-layer attacks that target software or cryptographic mechanisms, these attacks directly manipulate radio signals and transmission protocols, often causing significant disruptions in IoT communication [59]. The broadcast nature of wireless networks makes them particularly susceptible to such

threats, as adversaries can intercept, alter, or jam signals without requiring physical access to devices. Physical layer attacks can lead to Denial-of-Service (DoS), data leakage, and compromised network integrity. Some of the most common threats in this category are as follows:

- **Jamming Attacks:** Attackers introduce interference in the communication channel, preventing legitimate devices from transmitting or receiving data effectively [60,61]. Since IoT devices often operate in shared spectrum environments, such as Wi-Fi, Zigbee, LoRa, and Bluetooth, they are highly vulnerable to jamming. Attackers may emit high-power noise continuously, activate interference only when they detect legitimate transmissions, or use randomized noise patterns to evade detection. Jamming attacks are particularly effective against mission-critical IoT applications, including smart grids, industrial automation, and healthcare monitoring, where any disruption can lead to severe consequences.
- **Eavesdropping:** Passive attackers listen to unencrypted signals to capture sensitive information without being detected. This type of attack is especially dangerous when data is transmitted in plaintext or with weak encryption. Attackers may deploy high-gain directional antennas and software-defined radios to capture signals from long distances.
- **Spoofing and Man-in-the-Middle Attacks:** Malicious nodes impersonate legitimate devices, intercepting or altering transmitted data. Attackers may forge media access control addresses, signal characteristics, or cryptographic handshakes to trick devices into accepting them as trusted entities. Man-in-the-middle attacks extend this threat by not only intercepting communication but also modifying, injecting, or replaying messages in real time. This enables adversaries to alter IoT sensor data, inject false commands, or disrupt authentication processes, potentially leading to unauthorized device control or service manipulation.

To mitigate such risks, various physical layer security mechanisms have been developed. One notable approach is artificial noise injection, which involves adding controlled interference to wireless signals. This technique ensures that only the intended receiver, possessing the correct decryption key, can successfully interpret the message, while unauthorized eavesdroppers receive unintelligible signals. Advanced schemes dynamically adapt noise levels based on channel conditions and perceived threats [62,63]. Another promising strategy is compressive sensing, which enhances both privacy and efficiency by reducing the number of transmitted signal samples. Since attackers lack access to the full sampling structure, reconstructing the original data becomes computationally infeasible. This method is especially valuable in energy-constrained IoT environments where minimizing transmission overhead is crucial.

Bit flipping is also employed, whereby intentional errors are introduced into the transmitted data. Only the legitimate receiver, equipped with the proper key or decoding mechanism, can accurately restore the original message. This technique significantly diminishes the utility of passively intercepted communications and increases resistance to statistical analysis attacks [64]. In dense deployments such as smart cities or industrial IoT, cooperative secrecy mechanisms are utilized. These rely on collaboration between trusted nodes to enhance secrecy using techniques such as dummy transmissions, coordinated jamming, or data masking. By confusing adversaries while allowing legitimate communication to proceed, these methods offer distributed resilience to eavesdropping and traffic analysis [65–67]. Finally, physical-layer encryption leverages inherent wireless channel characteristics to generate dynamic, session specific cryptographic keys. This type of encryption often incorporates channel state information or RF fingerprints to create robust, unpredictable keys, protecting against brute-force decryption attempts and ensuring data confidentiality at the lowest communication layer [68,69].

Table 2
Security challenges in IoT architecture.

Challenge	Description
Resource constraints	IoT devices often have limited processing power, memory, and battery life, restricting the implementation of complex security algorithms [30–32].
Heterogeneous devices	The diversity of devices and manufacturers leads to inconsistencies in security capabilities and standards [33].
Scalability	Managing and securing a large number of devices in massive IoT deployments is challenging [34,35].
Physical vulnerability	Devices placed in unsecured locations are prone to physical attacks, such as tampering or node capture [36].
Communication protocols	Various protocols introduce vulnerabilities and complicate unified security measures [37].
Data privacy	Sensitive data transmission over IoT networks raises privacy concerns, especially in healthcare and smart homes [38].
Firmware and software updates	Lack of secure update mechanisms leaves devices vulnerable to exploitation through outdated software [39].
IoT botnets	Compromised devices can form botnets for Distributed Denial-of-Service attacks, exacerbated by weak authentication mechanisms [40–42].
Edge and fog computing integration	Decentralized data processing introduces new attack vectors and requires advanced security frameworks [43–45].
Dynamic network topologies	Changes in IoT network topology due to mobility or failures introduce vulnerabilities and complicate management.
Insider threats	Malicious actions by authorized users or compromised internal devices add another layer of complexity to security.
Post-Quantum security	Advances in quantum computing threaten traditional cryptographic methods, requiring quantum-resistant alternatives [46,47].
Adversarial machine learning	Attackers exploit vulnerabilities in machine learning models through adversarial inputs, model inversion, or poisoning attacks.
Interoperability issues	IoT ecosystems often face compatibility issues between devices from different manufacturers, creating potential security risks [48,49].
Supply chain attacks	IoT devices may be compromised during manufacturing or distribution through malicious software or hardware vulnerabilities [50,51].
Cloud integration risks	Connecting IoT devices to cloud systems introduces new vulnerabilities in both data security and cloud infrastructure [52].
AI-Powered treats	Attackers may use AI-driven techniques to automate attacks on IoT systems, making them harder to detect and mitigate.
Zero trust implementation	Traditional perimeter based security is inadequate for IoT; a zero trust approach is needed to continuously verify trust across users, devices, and services. This is challenging in resource-constrained or legacy environments.
Side-channel attacks	IoT devices are vulnerable to timing, power analysis, and electromagnetic attacks that extract sensitive data without breaching software defenses. Countermeasures include noise injection and secure hardware design.
Trust establishment and key management	Secure and scalable key distribution is challenging in dynamic, large-scale IoT systems. Traditional PKI is often too heavy, while lightweight alternatives (e.g., IBE, PSK) face interoperability or scalability issues.
Cross-layer risk propagation and dataset limitations	Most IDS studies isolate security challenges to individual IoT layers. Our analysis highlights how attacks propagate across layers from constrained devices to cloud systems and how cross-layer visibility improves detection granularity.
Challenges in federated and quantum-resilient IDS	Centralized IDS frameworks face inherent limitations such as privacy leakage, scalability bottlenecks, and vulnerability to single points of failure. While federated learning offers a promising alternative, it brings new challenges including data heterogeneity, increased communication overhead, model drift, and the need for decentralized trust. Simultaneously, the growing potential of quantum computing threatens classical cryptographic primitives and may render current IDS models obsolete, necessitating early research into quantum-resilient detection strategies despite limited existing work [53,54].
Blockchain based trust	Lack of decentralized coordination in IoT hinders secure trust management. Blockchain offers immutable logging, consensus based validation, and smart contract automation to resist insider threats and data tampering. Yet, resource constraints and protocol compatibility limit its widespread adoption [55,56].

2.1.2. Data link layer attacks

The data link layer in IoT networks manages frame transmission, error control, and media access coordination, ensuring reliable data transfer while handling contention, collision avoidance, and access scheduling [70–73]. However, its critical role in wireless communications makes it vulnerable to attacks that exploit weaknesses in protocols such as IEEE 802.15.4 (Zigbee, 6LoWPAN), Wi-Fi, LoRa, and Bluetooth

Low Energy [74,75]. These attacks can result in unauthorized access, frame manipulation, and Denial-of-Service, disrupting communication and compromising network integrity. Despite their known weaknesses, these protocols remain widely adopted in real-world IoT deployments. This is primarily due to their low energy consumption, simplicity, and support for resource-constrained devices. Protocols like IEEE 802.15.4 (used in Zigbee and 6LoWPAN), LoRa, and BLE were not originally

designed with strong security in mind, but they offer efficient communication for battery-powered or low-complexity nodes. Additionally, these protocols are often hardcoded into firmware or chipsets, making post-deployment security upgrades difficult or infeasible. As a result, the prevalence of these legacy protocols contributes significantly to the attack surface of IoT systems, especially in long-term industrial, environmental, or smart home deployments. One prominent example of such threats is jamming attacks, which intentionally interfere with legitimate communication by transmitting disruptive signals. These attacks can vary in form, including constant jamming, reactive jamming, and selective jamming, each differing in intensity and targeting key network frames, such as acknowledgments or routing updates [76–78]. Another related threat, Denial-of-Service (DoS), exploits vulnerabilities in frame transmission mechanisms, manipulating the Network Allocation Vector (NAV) timers or injecting fraudulent acknowledgment frames, further degrading network performance. Beyond interference based threats, attackers may directly compromise communication integrity through MAC spoofing and address manipulation. By impersonating legitimate devices, adversaries can gain unauthorized network access or execute man-in-the-middle attacks [79]. Frame replay attacks further exploit the lack of strong authentication mechanisms by retransmitting captured frames to bypass verification processes, disrupt data flow, or drain device resources. Additionally, collision attacks leverage channel contention to artificially increase retransmissions, degrade throughput, and exhaust battery life in resource-constrained devices. Wormhole attacks represent another significant threat by establishing unauthorized low-latency tunnels between distant nodes, misleading routing decisions, and enabling selective packet forwarding [80–82].

To mitigate these threats effectively, multiple countermeasures have been developed at the data link layer. Anti-jamming techniques, such as frequency hopping spread spectrum and direct sequence spread spectrum, dynamically alter transmission channels to avoid interference, while adaptive power control helps maintain stable communications under hostile conditions. Transitioning from contention based protocols like CSMA/CA to more secure MAC layer protocols such as TDMA can significantly reduce vulnerabilities related to backoff manipulation and NAV exploitation. Furthermore, incorporating cryptographic authentication and message authentication codes (MACs) provides an essential layer of defense against spoofing and unauthorized frame alterations. Intrusion detection systems can further enhance security by monitoring network anomalies, abnormal collision rates, and unauthorized MAC address modifications. Lastly, implementing delayed acknowledgment mechanisms introduces additional authentication checks, thereby reducing the risks associated with replay attacks.

2.1.3. Network layer attacks

Network layer attacks in IoT exploit vulnerabilities within communication protocols and routing mechanisms to disrupt device connectivity and compromise data integrity. A prominent threat at this layer includes DoS attacks, where adversaries flood networks with excessive traffic, consuming bandwidth and computational resources to degrade or halt legitimate communication. In more severe cases, Distributed Denial-of-Service (DDoS) attacks leverage multiple compromised IoT devices, known as botnets, to launch synchronized attacks, causing severe disruptions to critical systems such as smart grids, healthcare, and industrial automation. Another significant threat is the Man-in-the-Middle (MiTM) attack, in which attackers intercept and potentially manipulate communications between IoT devices without detection. These attacks are particularly critical in IIoT and smart city infrastructures due to their potential impacts on operational safety and data accuracy [83,84]. Attackers often exploit insecure protocols like MQTT or CoAP, intercepting, altering data transmissions, or injecting malicious commands. The continued use of insecure or legacy protocols at the network layer is not uncommon in IoT systems. Protocols such as RPL, AODV, or CoAP are lightweight and optimized for low-power, lossy networks, making them appealing for constrained devices.

However, their original designs often lacked strong authentication, encryption, or resistance to routing manipulation. Many devices are deployed in environments where firmware updates are rare or impossible, further prolonging the life of vulnerable protocol stacks. Consequently, the use of legacy routing and communication protocols remains a critical source of vulnerability in modern IoT deployments, requiring external protective mechanisms such as IDS or secure overlay protocols.

Due to their central role in facilitating communication among IoT devices, routing protocols such as RPL, AODV, OLSR, and DVR are particularly susceptible to targeted attacks that manipulate routing decisions. For instance, sinkhole attacks attract and intercept network traffic by falsely advertising optimal routes. Similarly, blackhole attacks advertise attractive paths but subsequently drop incoming packets, effectively isolating devices from the network. Wormhole attacks, though initially discussed in the context of the data link layer, also pose significant risks at the network layer. In this scenario, attackers establish unauthorized low-latency tunnels between distant nodes, misleading routing processes and potentially causing loops or packet loss. Selective forwarding and grayhole attacks degrade network reliability by selectively dropping specific packets, while rank attacks disrupt normal RPL operations by falsifying node rankings [85]. Additionally, replay attacks retransmit captured packets, triggering repeated or unauthorized network operations. Lastly, Sybil attacks, in which a single malicious node assumes multiple identities, undermine trust mechanisms, causing severe routing inconsistencies and instability. A detailed taxonomy of these routing attacks, categorizing their methodologies and impacts, is presented in Table 3 to better understand their operating mechanisms [86–88].

To mitigate these network layer threats, several protective measures are recommended. Secure and robust routing protocols can significantly enhance resistance against routing attacks. Network segmentation and traffic filtering help isolate compromised or sensitive devices, limiting attack spread. Implementing strong encryption mechanisms such as TLS or DTLS ensures confidentiality and integrity, effectively protecting against MiTM attacks. Trust based routing evaluations and secure rank computations further guard against rank and Sybil threats. Finally, employing intrusion detection and machine learning based anomaly detection systems can identify suspicious traffic patterns in real-time, providing rapid response capabilities essential for securing IoT networks [89,90].

2.1.4. Transport layer attacks

The transport layer plays a crucial role in enabling reliable end-to-end communication in IoT networks by supporting widely adopted protocols such as TCP and UDP. However, the inherent limitations of these protocols particularly when deployed over resource constrained IoT devices introduce several critical vulnerabilities. Adversaries often exploit these weaknesses to disrupt session integrity, exhaust system resources, or compromise data confidentiality. One of the most prevalent threats is SYN flooding, a denial-of-service attack wherein attackers send a flood of TCP SYN packets to initiate but not complete handshake processes [91]. These half open connections consume memory and processing resources on the target device, rendering it incapable of serving legitimate requests. In a similar vein, TCP Reset (RST) Injection attacks aim to abruptly terminate established TCP connections by injecting forged RST packets. This type of disruption is especially harmful to persistent IoT data streams, such as those used in telemetry or monitoring applications. A summary of major transport layer threats and corresponding mitigation strategies is provided in Table 4.

Another notable vulnerability is session hijacking, where attackers predict or intercept TCP sequence numbers to gain unauthorized control over an active session [92]. This facilitates data manipulation, injection of malicious commands, or forced termination of legitimate communication. To exacerbate the threat landscape, desynchronization attacks are used to disrupt the alignment between sender and receiver by forging packets that invalidate synchronization parameters.

Table 3

Categorization of attacks on RPL routing protocol [86–88].

Attack name	Attack methodology	Disadvantage
Resource based attacks		
Version number attack	The attacker modifies the version number to force frequent topology updates.	Increases energy consumption and destabilizes the network.
Flooding attack	The attacker sends excessive control messages to deplete network resources.	Reduces network efficiency and causes resource exhaustion.
Rank attacks	The attacker manipulates its rank to interfere with the routing path selection.	Leads to routing inconsistencies and increases packet loss.
Distributed denial-of-service	The attacker overloads the network with malicious traffic.	Causes network congestion and service unavailability.
DAO introduction attack	Fake DAO messages are sent to confuse the routing tables.	Increases control overhead and energy consumption.
Energy depletion attack	The attacker targets node energy resources through frequent message transmissions.	Shortens the lifespan of network nodes.
Local repair attack	The attacker forces unnecessary local repairs to waste energy.	Disrupts normal network operations and depletes resources.
Forwarding misbehavior attack	Nodes refuse to forward packets, interrupting the routing process.	Reduces the reliability of data delivery.
Hatchetman attack	The attacker selectively drops packets to disrupt the data flow.	Reduces delivery rates and network reliability.
Copypcat attack	The attacker clones the identity of a legitimate node to inject fake traffic.	Compromises security and trust in the network.
Topology based attacks		
Sinkhole attack	The attacker attracts network traffic to a compromised node.	Causes data loss and delays in communication.
Wormhole attack	The attacker creates a fake link between two distant nodes.	Disrupts routing decisions and decreases network performance.
Blackhole attack	Traffic is redirected to a malicious node, which then discards it.	Leads to significant data loss.
Selective forwarding attack	The attacker forwards only specific packets and drops others.	Results in incomplete data delivery and service disruptions.
Replay attack	The attacker replays old packets to confuse routing decisions.	Causes incorrect routing and resource wastage.
Greyhole attack	The attacker selectively drops packets based on certain conditions.	Decreases network reliability and efficiency.
Neighbor attack	The attacker manipulates neighbor tables by sending false information.	Creates routing loops and disrupts connectivity.
DODAG inconsistency	The attacker introduces inconsistencies in the DODAG structure.	Leads to communication breakdowns and inefficiency.
Advanced Vampire attack	The attacker generates unnecessary transmissions to drain energy.	Severely impacts the longevity of the network.
Divide and Conquer attack	The attacker splits the network into isolated segments.	Reduces the overall connectivity and efficiency of the network.
Tamper attack	The attacker alters control messages to mislead routing paths.	Compromises data integrity and routing accuracy.
Multiple mix attack	Combines several attack types to maximize disruption.	Significantly complicates detection and mitigation.
Ballot stuffing attack	The attacker inflates node rankings with fake votes.	Affects routing decisions and prioritization.
Poor mouthing attack	The attacker deliberately degrades its rank to mislead routing.	Causes sub-optimal routing paths and inefficiencies.
Opportunistic attack	Exploits specific vulnerabilities in routing protocols.	Reduces reliability and security of the network.
Traffic based attacks		
Sybil attack	A single attacker assumes multiple identities in the network.	Creates routing inconsistencies and trust issues.
Clone attack	The attacker duplicates legitimate nodes to introduce fake nodes.	Reduces data authenticity and security.
Decreased rank attack	The attacker manipulates its rank to attract data traffic.	Misleads routing and causes inefficient paths.
Forgery attack	The attacker forges data or control messages.	Misleads routing and causes resource wastage.
Eavesdropping attack	The attacker intercepts data packets to gather sensitive information.	Compromises data confidentiality and privacy.
Impersonation attack	The attacker pretends to be a legitimate node.	Reduces trust and disrupts network communication.

Table 4

Common transport layer attacks in IoT networks and mitigation strategies.

Attack	Protocol affected	Attack vector	Impact	Mitigation
SYN flooding	TCP	Incomplete TCP handshakes	Resource exhaustion	SYN cookies, rate limiting
Session hijacking	TCP	Sequence number prediction	Confidentiality breach	Sequence randomization, encryption
UDP reflection	UDP	Spoofed requests to public servers	DDoS amplification	Filtering, source validation
TCP RST injection	TCP	Forged RST packets	Session disruption	RST filtering, authentication
Energy drain	TCP/UDP	Repeated handshakes, spoofed resets	Battery depletion	Anomaly detection, watchdog timers
Desynchronization	TCP	Sequence/ACK manipulation	Communication instability	Packet authentication, session reset
Insecure protocol negotiation	CoAP/DTLS, MQTT/TLS	Devices accept weak or outdated protocol versions due to limited capabilities	Breaks encryption guarantees	Enforce minimum protocol standards, restrict legacy cipher suites

Table 5

Cross-layer taxonomy of IoT threats, protocol vulnerabilities, and defensive measures.

Layer	Attack types	Affected protocols/technologies	Attack vector/weakness	Security mechanisms/protocols	Limitations/Challenges
Physical	Jamming, Eavesdropping, Spoofing, MiTM	ZigBee, BLE, LoRa, Sub-GHz ISM, Wi-Fi, 6LoWPAN	- Open wireless medium with minimal access control - Absence of encryption in PHY/MAC headers - Susceptibility to radio interference and sniffing - Lack of physical tamper resistance in deployed IoT nodes - Inability to detect rogue devices or signal spoofing at the hardware level	- FHSS/DSSS modulation - Artificial noise injection - Physical-layer encryption/keyless secure transmission - RF fingerprinting for device authentication - PHY-aware IDS using spectral entropy and signal anomaly detection - Machine learning for jamming detection via SNR and RSSI anomalies	- Limited hardware support - Energy-constrained devices - Hard to distinguish between malicious and environmental interference
Data link	MAC spoofing, Replay, Wormhole, Collision	IEEE 802.15.4 (ZigBee, 6LoWPAN), LoRaWAN, BLE, TDMA	- No MAC-layer integrity/auth - Frame replay possible, Weak access control - Channel contention and backoff manipulation	- AES-128 link-layer encryption - MAC address filtering - Delayed ACK/TSCH (Time-Slotted Channel Hopping) - MAC-layer IDSs using timing and frame anomaly analysis - Cross-layer IDS using traffic patterns and timing anomalies - TSCH based secure MAC scheduling (IEEE 802.15.4e) - ML based anomaly detection on frame inter-arrival time and link-layer retries	- Lack of rekeying and key exchange protocols - Legacy protocols (e.g., IEEE 802.15.4, BLE, RPL) remain widely used in IoT due to firmware-level immutability, limited memory and processing capabilities, and long device lifecycles that prevent seamless adoption of modern secure protocols. - IDSs suffer from high false positive rate under bursty traffic - Requires packet inspection access at edge or gateway
Network	Sinkhole, Blackhole, Rank, Sybil, DAO Injection, DODAG Inconsistency, Wormhole, Replay	RPL, AODV, OLSRv2, CoAP, 6LoWPAN, ICMPv6, MQTT	- Insecure control messages, No source verification in routing, Poor trust propagation and loop avoidance, Neighbor Discovery attacks in 6LoWPAN-ND - Loop creation via version numbers or routing advertisements	- Secure RPL with pre-shared keys, IPSec, Lightweight Authentication and Secured Routing, Blockchain-secured RPL variants, Static Context Header Compression with security profiles, Trust based routing protocols, ML based IDS - Topology-aware anomaly detection for DODAG structure consistency	- IPSec/DTLS still too heavy - Blockchain increases latency and storage overhead - SCHC security mode support is limited in current hardware - ML based IDS requires training data and context awareness - Dynamic topologies hinder rule based anomaly detection - Lightweight but insecure protocols like RPL remain common due to constrained resources and lack of update mechanisms

(continued on next page)

The resulting retransmissions and protocol-level confusion can degrade system performance and reliability. For UDP based communications,

such as those used in CoAP, the lack of connection-oriented handshakes exposes devices to UDP flooding and amplification attacks. In the

Table 5 (continued).

Transport	SYN flooding, TCP RST Injection, Session hijacking, Desynchronization, Energy drain	TCP, UDP, CoAP over UDP, MQTT over TCP, DTLS	• Stateless nature of UDP limits session control • TCP handshake manipulation and sequence prediction • Lack of end-to-end authentication or secure session management • Absence of secure session establishment or integrity checks • High exposure to spoofed and reflection based DoS attacks	• DTLS (for UDP/CoAP), TLS (for MQTT/TCP) • TCP sequence randomization, Rate limiting and SYN cookies • Anomaly detection systems for traffic spikes and hijacking	• TLS handshake is costly • Edge/gateway offloading often required • Certificate management complex
Application	Firmware tampering, Malware injection, API abuse, Data poisoning, Cloud ransomware	CoAP, MQTT, REST APIs, HTTP, IoT Middleware, Cloud Services	• No default encryption/auth • Insecure API exposure, Poor firmware update security • Vulnerable cloud-side data processing and logging	• OSCORE, Secure MQTT • Signed OTA updates • OAuth 2.0, RBAC, MFA for access control • ML based behavioral profiling of app and firmware • Privacy-preserving behavioral analytics on device-cloud interactions • Integration of post-quantum cryptographic protocols (e.g., Kyber, Dilithium) for OTA updates and secure data exchange	• OSCORE still not widely adopted • Firmware behavior often undocumented, complicating IDS rule sets • Application-layer IDSs have higher false-positive rates • Cloud APIs inconsistently secured • Limited integration of post-quantum cryptography; current encryption relies on RSA/ECC, which are vulnerable to quantum attacks • No IDS validation for quantum-resilient protocol implementations

former, attackers overwhelm the device with high volumes of spoofed datagrams, while in the latter, they exploit open UDP services (NTP, DNS) to reflect and amplify traffic toward the victim using spoofed IP addresses. These attacks not only consume bandwidth but also deplete device-level processing capacity, often leading to service unavailability.

IoT devices are also susceptible to energy drain attacks, where adversaries continuously trigger protocol responses such as repeated handshakes or reset events to accelerate battery depletion [93]. This attack leverages the energy sensitivity of constrained devices to induce early failure or trigger system shutdowns in unattended environments. Mitigating transport layer attacks in IoT networks requires a multi-layered defense strategy. Proactive measures such as SYN cookies, rate-limiting, and connection timeout enforcement help reduce the impact of volumetric attacks. However, such measures are not commonly implemented directly on IoT devices due to their limited processing power, memory constraints, and energy sensitivity. Implementing computationally intensive defenses like SYN cookies or deep packet inspection can significantly affect the device's operational lifetime and real-time performance. As a result, these protections are more often deployed at the network gateway, edge devices, or cloud layers, where more robust resources are available to support complex security mechanisms. The deployment of secure transport protocols namely TLS for TCP based applications and DTLS for UDP based protocols enhances confidentiality and authenticity [94,95]. Additionally, randomized TCP sequence numbering, packet filtering, deep packet inspection, and anomaly detection systems can be employed to detect injection based and desynchronization attacks. Finally, energy-aware security policies and watchdog mechanisms should be implemented to detect and suppress abnormal packet activity that may lead to power exhaustion in IoT endpoints.

2.1.5. Application layer attacks

Although protocols like CoAP are widely adopted due to their efficiency in constrained environments, their default configurations typically do not include end-to-end security. To mitigate this, the IETF has standardized Object Security for Constrained RESTful Environments (OSCORE), which secures CoAP messages at the application layer by providing end-to-end encryption, integrity protection, and replay prevention even in the presence of intermediary nodes. OSCORE leverages

CBOR Object Signing and Encryption (COSE) to protect message payloads and options, ensuring that confidentiality and data authenticity are preserved across proxies or gateways. By integrating OSCORE, IoT systems can significantly reduce their exposure to application layer threats such as eavesdropping, message tampering, and unauthorized access, particularly in scenarios involving sensitive device control or data telemetry. One of the most severe threats at this layer is malware injection, where attackers introduce malicious code into IoT devices to gain control, exfiltrate data, or render devices inoperable [96]. Botnets such as Mirai and Mozi exploit weak authentication and outdated firmware, rapidly compromising large-scale IoT networks [97]. Similarly, code execution exploits target software vulnerabilities such as buffer overflows, insecure APIs, and improperly managed firmware updates, enabling attackers to manipulate device functionality or install backdoors [98].

IoT applications are also vulnerable to insecure API configurations, weak authentication tokens, and insufficient access controls. Attackers can exploit these weaknesses to manipulate transmitted data, execute unauthorized commands, or extract sensitive information [14]. Common threats include command injection attacks, cross-site scripting (XSS), and SQL injection [12,13]. Additionally, IoT cloud integrations introduce risks such as unauthorized API access and data poisoning, where adversaries manipulate transmitted data to corrupt analytics and machine learning models [99,100]. Data confidentiality and integrity are primary concerns at this layer. Eavesdropping attacks intercept unencrypted data exchanges, exposing credentials, encryption keys, or private user data [60,101]. Data exfiltration in cloud-integrated IoT environments is particularly dangerous, as attackers exploit weak access controls to extract confidential information, leading to privacy violations and operational risks. Additionally, data flooding attacks overwhelm network resources by injecting large volumes of false data, causing denial-of-service conditions [102]. Firmware security is another critical concern, as attackers frequently exploit vulnerabilities in update mechanisms to inject malicious firmware or introduce unauthorized modifications [16]. Devices lacking secure update protocols are particularly at risk, allowing adversaries to install persistent threats such as rootkits or disable security features [17,18].

To mitigate these risks, manufacturers should implement cryptographically signed firmware updates, secure boot mechanisms, and real-time integrity verification [19].

Beyond firmware vulnerabilities, application-layer threats extend to cloud environments, where IoT-generated data is processed and stored. Multi-tenant cloud platforms introduce risks such as unauthorized API access, insecure identity management, and insufficient encryption mechanisms [52]. Attackers can exploit these vulnerabilities to gain unauthorized control over IoT cloud infrastructures, manipulate system configurations, or disrupt services. One of the most concerning threats in cloud based IoT ecosystems is ransomware, where adversaries encrypt critical system files and demand ransom payments [103–105]. Given the reliance of industrial, healthcare, and smart infrastructure applications on IoT cloud services, such attacks can have significant operational and financial consequences. To mitigate these application-layer threats, IoT networks require a multi-layered security approach. Role based access control (RBAC) and multi-factor authentication (MFA) help prevent unauthorized access to IoT applications [40, 41]. Secure over-the-air (OTA) update mechanisms should be implemented to ensure that firmware and software patches are verified before installation [42]. Anomaly detection techniques can enhance security by identifying irregular patterns in API requests, device access, and data flow [43,44]. Furthermore, robust encryption protocols such as Transport Layer Security (TLS) and Secure MQTT (SMQTT) are essential to safeguarding data transmissions against interception and manipulation [45]. However, many of these application-layer encryption mechanisms rely on public-key cryptographic primitives such as RSA and ECC, which are mathematically vulnerable to quantum algorithms like Shor's algorithm [106,107]. Once scalable quantum computers become available, these algorithms can be broken in polynomial time, compromising the confidentiality and authenticity of critical data transmissions [108]. This is particularly concerning for long-lived IoT deployments such as industrial control systems or healthcare devices, which may remain operational for decades without frequent firmware upgrades. Despite rising interest in post-quantum cryptography (PQC), current IDS frameworks seldom integrate quantum-resistant algorithms such as Kyber or Dilithium both of which are part of the NIST PQC standardization effort [109–111]. Furthermore, IDS models do not typically consider the implications of quantum-era threats on OTA authentication, encrypted cloud telemetry, or secure boot mechanisms. Future application layer IDS designs should incorporate PQ-safe cryptographic schemes where feasible and include threat models that account for adversaries equipped with quantum capabilities [53]. In parallel with cryptographic and security mechanisms, blockchain technology has emerged as a promising approach to application-layer IDS in IoT systems [112]. Its decentralized and tamper-proof ledger enables secure sharing of threat intelligence and audit logs across heterogeneous devices without reliance on a central authority [113,114]. Smart contracts can automate incident response policies, trigger firmware rollback, or revoke device access based on real-time anomaly alerts. Moreover, federated learning frameworks combined with blockchain have been proposed to collaboratively train detection models across edge devices while ensuring data privacy and traceability. Despite these advantages, blockchain based IDSs face deployment challenges including consensus overhead, integration with lightweight IoT protocols, and smart contract execution costs. Nevertheless, incorporating blockchain mechanisms into the application layer can significantly improve trust, traceability, and automation in intrusion detection for large-scale environments [115]. In addition to threat intelligence sharing, blockchain based IDS architectures can enforce decentralized identity management, secure access control, and immutable forensic logging capabilities that are critically needed at the application layer where trust boundaries are often weakest. These mechanisms can be integrated with IoT communication protocols such as CoAP, MQTT, and REST APIs through side channel validation, secure payload anchoring strategies.

This allows verifiable and tamper-proof detection logs to be generated across distributed devices without reliance on a central authority.

As IoT continues to expand across various industries, ensuring robust security at the application layer is paramount. A proactive approach that integrates strong authentication, secure firmware management, encrypted communications, and intelligent threat detection will be key to safeguarding IoT ecosystems against emerging cyber threats [52].

3. Intrusion detection methods for IoT networks

IDS play a vital role in securing IoT networks by identifying and mitigating cyber threats. As IoT environments become increasingly complex, traditional security mechanisms often fall short in detecting sophisticated attacks. IDS address this gap by monitoring network traffic, device behavior, and communication patterns to detect anomalies and known attack signatures, thereby preventing unauthorized access and service disruptions [116]. The unique characteristics of IoT networks such as their distributed architecture, limited resource availability, and reliance on heterogeneous communication protocols introduce significant security challenges. Consequently, various IDS methodologies have been developed, including signature, anomaly, specification, and hybrid approaches, each offering specific advantages [117]. These methodologies are visually summarized in Fig. 5. The integration of machine learning and deep learning techniques has further enhanced the accuracy and adaptability of these systems [118]. In this section, a comprehensive review of IDS research for IoT networks is presented, encompassing both general review articles and specific detection methods proposed in the literature. While survey based studies provide valuable insights into the evolution, strengths, and limitations of IDS technologies, proposed detection methods offer novel solutions tailored to the constraints and dynamics of IoT environments. By categorizing and analyzing these two strands of research, this section aims to offer a balanced overview of the field, capturing both prevailing trends and cutting-edge developments. As illustrated in Fig. 6, the growing number of IDS-related publications from 2016 to 2024 underscores the increasing significance and academic interest in this domain.

3.1. Review articles on IDS for IoT

This subsection highlights review articles that provide comprehensive surveys of IDS techniques for IoT networks. Articles were collected from major academic databases including IEEE Xplore, Elsevier, Springer, Wiley, MDPI, ACM Digital Library, and Taylor & Francis. The search query “IoT and IDS and Review Article” was used to ensure coverage of IDS methodologies and IoT security challenges. These studies classify IDS approaches while analyzing their scalability, accuracy, and efficiency. Additionally, they emphasize the difficulties of deploying IDS in resource-constrained IoT environments, such as limitations in processing power, memory, and energy.

Liao et al. provide a comprehensive review of IDS methodologies, categorizing them into signature based and anomaly based models while analyzing their effectiveness in mitigating modern cyber threats [121]. The study also discusses IDS applications in virtualized environments, addressing security challenges in cloud computing and virtual machines. Hajiheidari et al. conduct a systematic review of IDS mechanisms for IoT, categorizing approaches into anomaly based, signature based, specification based, and hybrid techniques [122]. Their study emphasizes IDS implementation challenges in resource constrained environments, evaluating detection accuracy, scalability, and computational efficiency. Arshad et al. examine the trade-offs between detection accuracy, energy consumption, and computational overhead in IDS models for IoT [123]. The study categorizes IDS approaches while identifying critical challenges, including the lack of standardized evaluation metrics and the need for real-world datasets to

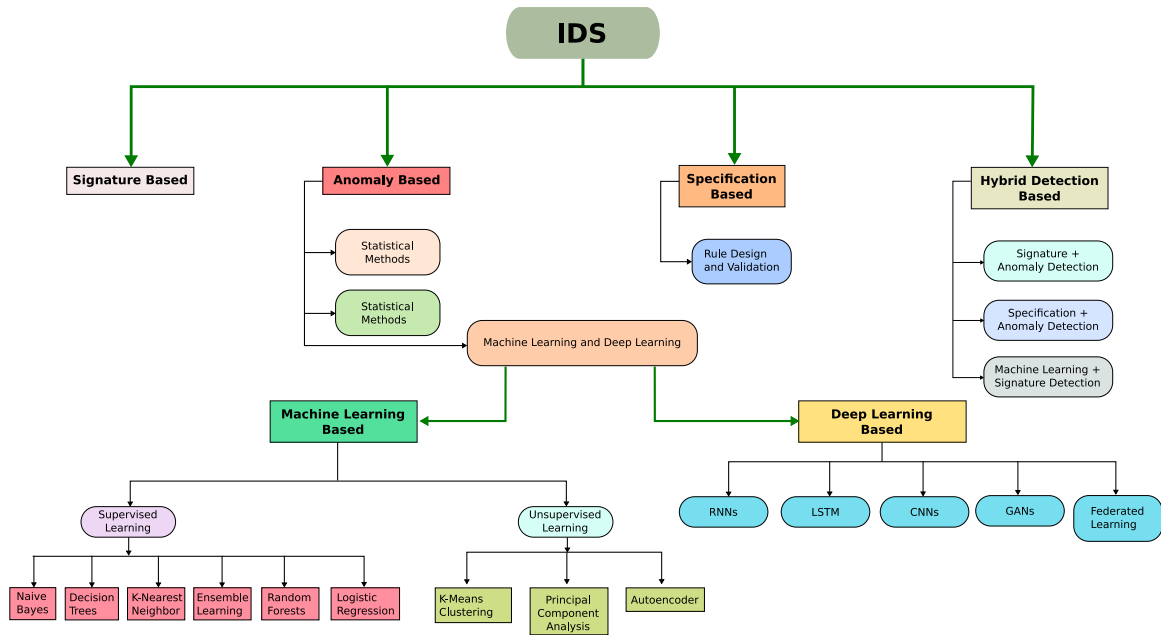


Fig. 5. Taxonomy of IDS categorized into signature based, anomaly based, specification based, and hybrid detection methods [117,119,120].

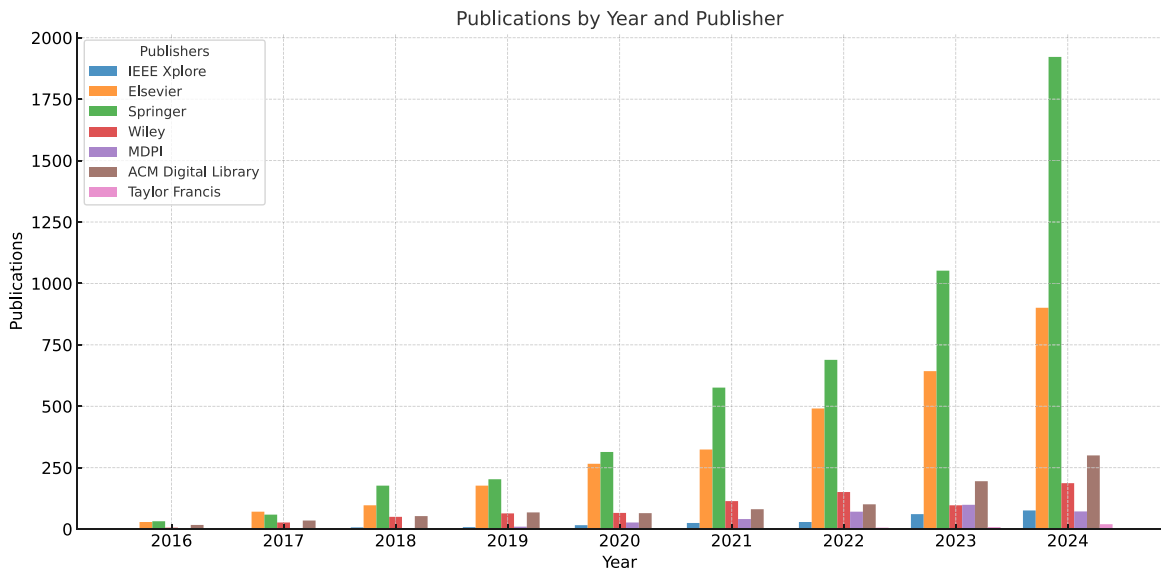


Fig. 6. Number of research on IDS in IoT (2016–2024) across major academic platforms.

improve IDS performance and applicability. Dalipi and Yayilgan survey the security and privacy challenges of integrating IoT technologies into smart grids, focusing on the customer, communication, and grid domains [124]. They highlight risks such as identity theft, behavior profiling, and data misuse at the customer level, alongside communication vulnerabilities like spoofing, eavesdropping, and DDoS attacks. SCADA systems and smart meters within the grid domain are noted for their susceptibility to physical tampering and software flaws. Similarly, Irshad's systematic review examines IoT security frameworks in terms of governance, risk management, and architectural coverage [125]. The study compares several frameworks, showing that infrastructure-level solutions offer features like secure boot and intrusion detection, whereas application-layer methods focus on lightweight security suitable for constrained devices. Irshad concludes that a comprehensive, unified governance model that integrates security across the device, network, and application layers is still absent in current IoT security approaches.

Abdulganiyu et al. present a systematic review of network IDS, focusing on anomaly based, signature based, and hybrid approaches [126]. Their study, structured using the PRISMA methodology, categorizes IDS techniques while analyzing datasets and performance metrics. The research highlights the transition from traditional signature based detection to anomaly based methods, with machine learning and deep learning playing an increasingly dominant role. Similarly, Wu et al. provide an extensive review of deep learning based IDS, categorizing techniques such as autoencoders, deep belief networks, convolutional and recurrent neural networks, generative adversarial networks, and transformers [127]. Their study discusses the advantages of deep learning in intrusion detection while identifying key challenges, including dataset limitations, feature selection complexities, and generalization issues. Bansal and Singhrova examine IDS methodologies in IoT and IIoT, categorizing detection techniques into signature and anomaly based models [128]. Their work highlights evolving security challenges, emphasizing the importance of improving detection accuracy,

computational efficiency, and adaptability to new threats. Albulayhi et al. contribute by providing a comprehensive taxonomy and reference architecture for anomaly based intrusion detection using deep learning, benchmarking performance across datasets such as BoT-IoT and IoTID20 [129]. Their study evaluates supervised, unsupervised, and hybrid learning models, identifying gaps in dataset diversity and real-world applicability. Similarly, Khraisat et al. analyze IDS deployment strategies, discussing the trade-offs between signature based, anomaly based, and hybrid IDS models in resource-constrained IoT environments [130]. Alsoufi et al. conduct an extensive review of deep learning techniques for anomaly based IDS, emphasizing their effectiveness in detecting zero-day attacks while addressing challenges in real-time threat detection and resource constrained environments [131]. Their study classifies IDS methodologies into supervised, unsupervised, and semi-supervised learning models, evaluating them across key performance metrics such as accuracy and false positive rates. However, they identify limitations in existing datasets, noting that many are outdated and fail to reflect modern IoT network traffic. Furthermore, they suggest that while deep learning methods enhance detection capabilities, improvements are needed in computational efficiency and time complexity to support real-world deployment. Adhikari et al. focus on anomaly detection in IoT, categorizing approaches into statistical, machine learning, deep learning, and hybrid methods [132]. They emphasize the need for scalable IDS solutions capable of adapting to real-time IoT network dynamics, along with the importance of robust datasets and evaluation frameworks for enhancing IDS reliability. Complementing this perspective, Zarpelão et al. propose a taxonomy of IDS approaches anomaly based, signature based, and classify IDS deployments as local, distributed, or hybrid [133]. Their study highlights gaps in attack coverage and validation methods, particularly in relation to the placement of IDS within diverse IoT environments. Expanding on architectural considerations, Benkhelifa et al. critically examine IDS designs in IoT, focusing on the challenges posed by protocol heterogeneity and resource constraints [134]. They categorize IDS architectures into centralized, distributed, hierarchical, and hybrid models, and discuss their suitability for handling varied communication protocols such as WiFi, RF, and 6LoWPAN. Sicato et al. provide a comprehensive analysis of IDS solutions for IoT, proposing a software-defined networking based distributed cloud IDS architecture designed to enhance detection accuracy and scalability [135]. Experimental evaluation using the NSL-KDD dataset yielded promising results, with accuracy ranging from 60% to 97% across various attack classes, such as DoS, U2R, and R2L. The study further addresses key challenges including data privacy, trust management, secure alert transmission, and scalability in large-scale IoT networks. Ozkan-Okay et al. present a comprehensive systematic review of IDS technologies, methodologies, detection approaches, datasets, and tools [136]. The study categorizes IDS into network based, host based, wireless, and network behavioral analysis, detailing their strengths, limitations, and detection capabilities. It emphasizes challenges in addressing dynamic cyber threats, outdated datasets, and the need for standardized evaluation metrics. The authors suggest hybrid IDS models and improved datasets as future directions. While the review offers broad coverage, it lacks empirical evaluations, which limits direct insights for practical deployments.

Ayyagari et al. present a systematic review of intrusion detection techniques in network environments [137]. The study evaluates signature based, anomaly based, and hybrid detection approaches, highlighting their advantages and limitations. It emphasizes the role of machine learning in improving IDS efficiency but notes challenges such as high false positive rates and computational overhead. Mishra and Pandya's study systematically reviews IoT security challenges, DDoS attacks, and IDS [138]. The paper effectively highlights the multi-layered vulnerabilities in IoT ecosystems and provides a thorough analysis of existing IDS techniques and their applications. The focus on DDoS attacks and anomaly detection is particularly valuable, given the increasing prevalence of such threats in IoT environments. The

study analyzes intrusion detection techniques in detail and includes detailed discussions on dataset use, machine learning, and deep learning methodologies. However, the study has some limitations. While it provides an extensive review, the reliance on outdated datasets limits the generalizability of its conclusions for modern IoT applications. Additionally, practical implementation insights and real-world examples could have enhanced the study's applicability. Expanding on these foundations, Heidari presents a systematic and detailed review of intrusion detection systems tailored for IoT, analyzing research published between 2018 and 2022 [117]. They further examine placement strategies such as centralized, distributed, and hybrid deployments, highlighting how these affect detection performance and system overhead in resource-constrained IoT environments. A distinctive contribution of the study is its critical assessment of widely used datasets like NSL-KDD, KD-99, CTU13, UNSW-NB15, and BoT-IoT, as well as the simulation tools employed across the reviewed works. Addressing the gap in data resources highlighted by these prior studies, Keersmaecker et al. conduct an extensive review of publicly available IoT datasets for network security research, surveying over 100 articles and 74 datasets [139]. Their taxonomy aids researchers in selecting appropriate datasets for specific IDS research needs and exposes significant deficiencies in dataset diversity, realism, and accessibility.

Alotaibi provides a comprehensive survey on security challenges, threats, and AI based solutions in IIoT environments [140]. The study categorizes IIoT security threats into three main layers perception, network, and application and highlights security requirements such as confidentiality, integrity, and availability. The paper further discusses how AI and edge/fog computing can enhance IIoT security by mitigating cyber threats and optimizing resource utilization. While the survey effectively identifies key challenges and solutions, it lacks empirical validation of AI techniques in real-world IIoT settings, limiting its applicability. Alotaibi provides a broad review of security challenges, threats, and AI based solutions in IIoT environments. The study categorizes IIoT security threats into three layers: perception, network, and application, emphasizing key security requirements like confidentiality, integrity, and availability. AI techniques and edge/fog computing are discussed as mechanisms to enhance IIoT security by mitigating cyber threats and optimizing resource utilization. Authors review recent advancements in artificial intelligence for enhancing intrusion detection in IoT networks [141,142]. Their work categorizes AI based IDS techniques, including ML, DL, and evolutionary computing approaches. The study highlights the strengths of AI in identifying novel attacks and adapting to evolving threats, particularly through anomaly detection and hybrid models. However, the review also emphasizes challenges such as dataset standardization, model generalizability, and computational overhead, which limit the practical deployment of AI based IDS in resource-constrained IoT environments. Roy et al. review energy-efficient IDS solutions for IoT, focusing on optimizing security mechanisms for resource-constrained devices [143]. Their study categorizes IDS techniques based on strategies like lightweight anomaly detection, adaptive sampling, and event-driven data processing, aiming to balance energy efficiency with detection accuracy. While the study proposes a framework for sustainable IDS implementations, it does not extensively explore hybrid IDS architectures and lacks experimental validation of the proposed energy-saving methods.

Al-Haija and Droos present a comprehensive survey on deep learning based IDS in IoT environments [144]. Their study categorizes existing IDS methodologies, highlighting the advantages of deep learning in detecting complex attack patterns and adapting to dynamic IoT threats. The authors emphasize the scalability and adaptability of deep neural networks in intrusion detection while also discussing challenges such as data scarcity, computational overhead, and adversarial robustness. The survey reviews benchmark datasets and suggests future research directions, particularly in enhancing real-time IDS performance and developing more efficient learning models. Halvorsen et al. provide a systematic mapping study and review on the application

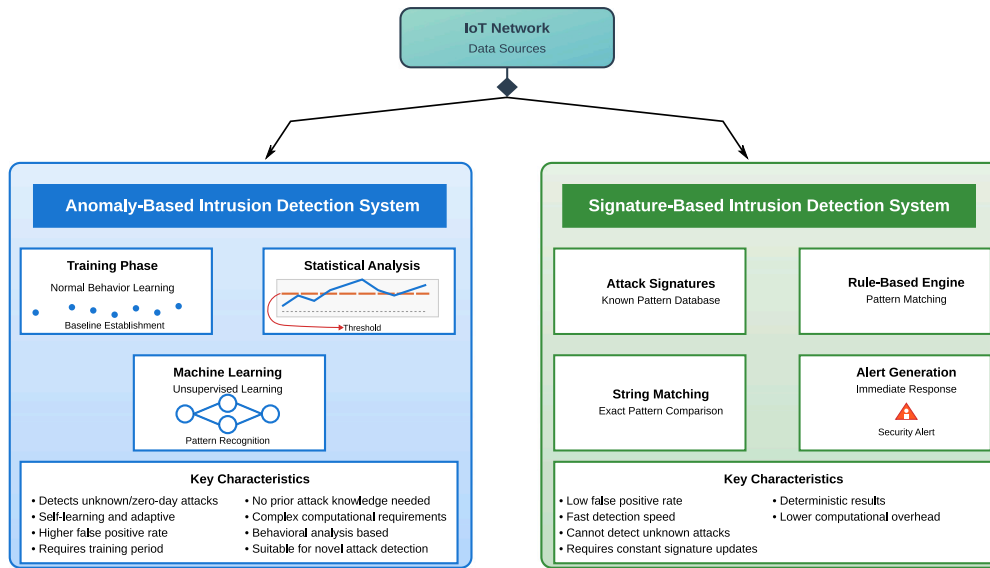


Fig. 7. Conceptual illustration of anomaly based vs. signature based intrusion detection systems [152].

of Generative Machine Learning Models (GMLMs) in IDS [145]. The study categorizes research into three main areas: penetration testing, dataset augmentation, and standalone IDS implementations using GMLMs. The authors highlight that while GMLMs enhance anomaly detection and mitigate dataset limitations, they also face challenges such as stability in training and evaluation standardization. The paper critically evaluates GANs, VAEs, and diffusion models, discussing their role in improving IDS resilience against novel attacks. Kikissagbe and Adda provide a comprehensive review of intrusion detection methods in IoT systems, focusing on supervised, unsupervised, and deep learning approaches [146]. The study assesses the effectiveness, limitations, and practical applications of these techniques, highlighting their role in enhancing IoT security. The authors emphasize the challenges posed by the heterogeneity and scalability of IoT environments, as well as the need for adaptable IDS models.

Dritsas and Trigka examine the role of cloud computing in IIoT security, discussing how cloud based frameworks enhance real-time processing, predictive analytics, and anomaly detection [147]. Their study emphasizes AI-driven security solutions, including blockchain and IDS integration, to strengthen IIoT networks. Additionally, they explore future research directions such as 5G, serverless computing, and federated learning for more efficient intrusion detection and prevention. Walling and Lodh review ML and DL based IDS, comparing their effectiveness based on datasets, validation strategies, and performance criteria [148]. Their study highlights ML's efficiency in resource constrained environments and DL's capability to detect zero-day attacks but notes challenges in scalability, real-time performance, and adaptability to real-world IoT scenarios. Mohale and Obagbuwa analyze the application of Explainable Artificial Intelligence (XAI) in IDS, categorizing methods such as SHAP, LIME, rule based models, and decision trees [149]. Their study underscores the trade-offs between interpretability and detection accuracy, noting that while rule based and decision-tree models offer transparency, they may reduce detection performance. Key challenges include balancing accuracy with interpretability and mitigating the risks of exposing sensitive security insights. Ahsan et al. introduces a detailed taxonomy of metaheuristics, categorizing them into swarm-based, nature-inspired, evolutionary, and physics/math-based algorithms, and examines their applications in feature selection, parameter optimization [150]. Additionally, the study identifies critical challenges such as dataset imbalance, resource constraints in IoT devices, and the lack of real-world validations. Rahman et al. develop a structured taxonomy covering ML, DL, ensemble, and

hybrid models, alongside a critical analysis of datasets and evaluation metrics [151]. Notably, their analysis reveals that ensemble methods (e.g., Random Forest and AdaBoost) achieved accuracies of up to 99.42% on the WSN-DS dataset and 97.77% on UNSW-NB15. Meanwhile, deep learning models like LSTM and CNN reached 99% accuracy on both the BoT-IoT and ToN-IoT datasets.

Tables 6 and 7 present a critical and comparative analysis of existing review articles on IDS for IoT, detailing each study's primary focus, methodological contributions, identified research gaps, and noted limitations. To underline the novelty of this survey, our work has also been added to the tables, emphasizing its cross-layer taxonomy, its critical discussion of recent IoT-specific datasets, and the inclusion of an empirical case study on RPL based routing attacks using the ROUT-4-2023 dataset. In this way, the distinct contributions of this manuscript such as addressing constrained wireless environments, highlighting future challenges like post-quantum threats and federated learning, and offering a dataset-grounded perspective with empirical validation are made explicit in direct comparison with prior surveys.

3.2. Signature based detection in IoT

Signature based intrusion detection identifies threats by comparing observed network activity to a database of known attack signatures [153]. This method is highly effective for detecting well-documented threats with minimal false positives. However, its reliance on predefined signatures limits its ability to detect novel or evolving cyber threats, including zero-day attacks [154]. Maintaining an up-to-date signature database is also resource intensive, posing challenges for IoT devices with constrained processing and memory resources. Despite these limitations, signature based detection remains a critical component of IDS, particularly when integrated with anomaly based or machine learning techniques. Fig. 7 offers a simplified conceptual view of signature based and anomaly based intrusion detection systems; a detailed IoT-specific analysis is provided in the subsequent tables.

In recent years, numerous studies have sought to enhance the capabilities of signature based IDS, especially in IoT environments. Yuxin et al. propose a hash based contextual signature approach to improve efficiency by reducing non-critical alarms [155]. Their method integrates contextual information with IDS signatures and utilizes hash functions to develop an adaptive filtering mechanism, minimizing false positives and irrelevant alerts. Meharouech et al. propose a network security alert management architecture for NAT environments, enhancing attack attribution and reducing redundant notifications through

Table 6

Review articles on IDS for IoT.

Year	Authors	Primary focus/scope	Methodological contribution	Identified gaps/future directions	Noted limitations
2013	Liao et al. [121]	General IDS methodologies and their application in virtualized environments.	Provided a foundational taxonomy of IDS, categorizing them into signature and anomaly based models.	Addressed emerging security challenges in cloud computing and virtual machines.	Lacks empirical performance evaluations and an IoT-specific focus.
2016	Dalipi and Yayilgan [124]	Security and privacy challenges for IoT applications in smart grids.	Offered a structured analysis of threats across customer, communication, and grid domains.	Highlighted risks like identity theft, data leakage, and system vulnerabilities in SCADA and smart meters.	Narrow focus on the smart grid vertical; does not cover general IoT IDS methodologies.
2016	Irshad [125]	Systematic review of information security frameworks in IoT with a focus on governance and risk management.	Provided a comparative analysis of security frameworks (e.g., Floodgate, CoAP) based on policy integration and risk governance.	Concluded that a comprehensive, unified governance model integrating security across all IoT layers is still absent.	Conceptual focus on frameworks rather than technical IDS implementations.
2017	Zarpelão et al. [133]	Survey of early IDS research efforts specifically for the IoT paradigm.	Proposed a comprehensive, IoT-centric taxonomy based on detection method, placement, threats, and validation.	Pointed out significant gaps in attack coverage and the validation strategies used in early IoT IDS research.	Primarily a conceptual survey; lacks analysis of ML/DL advancements that followed.
2018	Benkhelifa et al. [134]	Critical examination of IDS architectures and methods for heterogeneous IoT environments.	Categorized IDS architectures (centralized, distributed, etc.) and assessed their suitability for constrained, multi-protocol IoT networks (RF, WiFi, 6LoWPAN).	Emphasized the limitations of traditional IDS in handling IoT's technological diversity and resource constraints.	Focuses on architectural challenges rather than a deep dive into specific detection algorithms or datasets.
2019	Hajiheidari et al. [122]	Systematic review of IDS techniques for IoT with a focus on scalability.	Provided a structured taxonomy of IDS approaches (anomaly, signature, specification, hybrid).	Focused on the challenges of implementing IDS in resource-constrained environments.	Lacks performance validation across diverse IoT architectures and relies on early-stage literature.
2020	Arshad et al. [123]	Performance, energy, and privacy trade-offs in ML based IDS for IoT.	Presented an extensive taxonomy and evaluation of IDS approaches from a resource-efficiency perspective.	Identified the lack of standardized evaluation metrics and the need for realistic, real-world datasets.	Lacks empirical validation and experimental benchmarks to substantiate its analysis.
2020	Sicato et al. [135]	In-depth analysis of IDS for IoT, proposing a novel SDN based distributed cloud IDS framework.	Introduced a software-defined IDS framework using SDN and ML, validated on the NSL-KDD dataset.	Addressed challenges including data privacy, trust management, secure alert transmission, and scalability.	Reliance on the outdated NSL-KDD dataset limits the relevance of its findings to modern IoT threats.
2021	Ozkan-Okay et al. [136]	Comprehensive systematic review of IDS technologies, methodologies, datasets, and tools.	Provided a holistic evaluation of IDS types (network based, host based, wireless, NBA) and their strengths/limitations.	Suggested hybrid IDS models and improved datasets as future directions to address dynamic cyber threats.	Offers broad coverage but lacks empirical evaluations, limiting direct insights for practical deployments.
2021	Alsoufi et al. [131]	Systematic review of anomaly based IDS in IoT using Deep Learning (DL) techniques.	Classified DL based IDS into supervised, unsupervised, and semi-supervised models and evaluated their metrics.	Highlighted the need for improved computational efficiency and reduced time complexity for real-world deployment.	Noted that many existing datasets are outdated and do not reflect modern IoT network traffic.
2021	Ayyagari et al. [137]	Systematic review of intrusion detection techniques in general network environments.	Offered a structured taxonomy of IDS approaches and explored ML-driven advancements.	Emphasized the role of ML in improving IDS efficiency but noted challenges like high false positives.	Does not analyze the specific deployment complexities of these techniques in IoT settings.
2021	Albulayhi et al. [129]	Taxonomy and reference architecture for anomaly based IDS using DL.	Benchmarked performance of supervised, unsupervised, and hybrid models on modern datasets (BoT-IoT, IoTID20).	Identified gaps in dataset diversity and the need for validation in real-world application scenarios.	The proposed reference architecture and analysis lack real-world validation.
2021	Mishra and Pandya [138]	Systematic review of IoT security vulnerabilities, DDoS attacks, and IDS.	Provided a thorough analysis of existing IDS techniques with detailed discussions on dataset use and ML/DL methodologies.	Offered actionable insights into IoT security and directions for future advancements.	Reliance on outdated datasets limits the generalizability of its conclusions for modern IoT applications.
2021	Khraisat et al. [130]	Survey on IDS methodologies, detection strategies, and validation techniques for IoT security.	Provided a structured analysis of IDS approaches (signature, anomaly, hybrid) in resource-constrained contexts.	Discussed the trade-offs between different IDS models for IoT environments.	Lacks empirical validation in real-world IoT deployments.

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NAT-aware intrusion detection [156]. Mutep et al. introduce an auto-updating signature based IDS model, dynamically structuring the signature database to optimize query execution and reduce system overhead [157].

Furthermore, Schrötter et al. evaluate deep-learning based IDS against traditional signature based systems in IoT networks, finding that signature based IDS such as Snort remained more reliable for detecting known patterns [158]. Hybrid approaches have also emerged;

Table 6 (continued).

2022	Heidari and Jamali [117]	Systematic review of IoT IDS research published between 2018 and 2022.	Provided a critical assessment of widely used datasets (NSL-KDD, UNSW-NB15, Bot-IoT) and simulation tools.	Analyzed the impact of different placement strategies (centralized, distributed, hybrid) on performance and overhead.	Lacks empirical validation and practical deployment examples; primarily synthesizes existing literature without proposing concrete IDS frameworks or performance evaluations.
2023	Keersmaeker et al. [139]	Comprehensive review of publicly available IoT datasets for network security research.	Surveyed over 100 articles and 74 datasets, creating a taxonomy to aid researchers in dataset selection.	Exposed significant deficiencies in existing datasets regarding diversity, realism, and accessibility.	Focuses on datasets rather than IDS methodologies themselves.
2023	Alotaibi [140]	Survey on security challenges, attacks, and AI based solutions in IIoT environments.	Categorized IIoT security threats by layer and discussed how AI and edge/fog computing can enhance security.	Highlighted the role of edge and fog computing in mitigating cyber risks.	Lacks empirical validation of the discussed AI techniques in real-world IIoT settings.
2024	Abdulganiyu et al. [126]	Systematic review of anomaly, signature, and hybrid based IDS methods using PRISMA.	Highlighted the transition from signature based to ML/DL-driven anomaly based methods.	Identified gaps in hybrid IDS research and re-emphasized the need for updated datasets and real-time deployment solutions.	Focuses on general network IDS with some IoT context, but not exclusively on IoT challenges.
2024	Bansal and Singhrova [128]	Review of IDS methodologies in IoT and IIoT.	Categorized detection techniques into signature and anomaly based models.	Highlighted evolving security challenges and the need for improved accuracy, efficiency, and adaptability.	Discusses evolving approaches but lacks empirical validation.
2024	Saied et al. [141]	Review of AI-driven IDS techniques for IoT security.	Categorized AI based IDS techniques, including ML, DL, and evolutionary computing approaches.	Proposed AI based adaptive IDS solutions to address evolving threats.	Emphasized challenges in dataset standardization, model generalizability, and computational overhead.

Otoun et al. propose a hybrid anomaly and signature based IDS (AS-IDS) utilizing Deep Q-learning for detecting unknown threats while maintaining signature based detection for known ones [159]. Similarly, Sajad et al. integrate Suricata IDS with neural networks and fuzzy logic to create a hybrid detection framework achieving 96.11% accuracy [160]. Meta-learning strategies have also been explored. Umer et al. present Meta-IDS, a meta-learning based hybrid system for IoMT networks, achieving high accuracy and execution speed [161]. Game theoretic models, such as those proposed by Basant et al. and Punithan et al. have optimized intrusion signature configurations to reduce false alarms and improve detection accuracy [162,163].

Emerging technologies like blockchain have also been leveraged, with Wenjuan et al. proposing CBSigIDS, a blockchain based collaborative IDS framework to secure signature sharing in distributed IoT networks [164]. Other advancements include using MapReduce frameworks to parallelize pattern matching and enhance performance, as demonstrated by Aldwairi et al. [165]. Finally, studies such as those by Amit et al. have emphasized that machine-learning based IDS models outperform traditional signature based systems when handling emerging threats, reinforcing the need for adaptive and intelligent intrusion detection frameworks in IoT environments [166]. Building on the advancement of protocol-specific IDS, Middlesex et al. introduce a signature based IDS tailored for the AMQP within IoT networks [167]. They develop a comprehensive testbed simulating real-world AMQP deployments to generate datasets encompassing normal, malicious, and mixed traffic. The model leverages traffic features such as TCP flags, sequence numbers, and payload sizes to define unique attack signatures. Experimental evaluations demonstrate high detection accuracy, including 71.5% malicious traffic identification for DoS, over 63% detection for MiTM, and over 90% for brute force attempts, all with minimal false positives on benign traffic. Focusing on wireless networks, Thankappan et al. propose a specialized signature based IDS designed to detect advanced Multi-Channel MitM attacks targeting protected Wi-Fi networks, which are prevalent in IoT environments [168]. The system achieves a minimum true positive rate TPR of 90% at short-range detection and 84% at 12 m, with an average detection time of 50–55 s using a 60-s probe interval. Moreover, the authors contribute to the research community by publishing a novel dataset containing traffic

traces of MC-MitM attacks, which was previously unavailable. Alnasser et al. introduced SBAIDS, an IDS framework that combines decision tree based signature detection with LSTM based anomaly detection to secure IoT and V2G networks [169]. The system achieve 95.6% accuracy and 93.9% F1-score when evaluated on enhanced versions of the CSE-CIC-IDS2018 and CIC-DDoS2019 datasets. A key feature of the framework is its joint detection database, which reduces false positives by 27% by integrating both detection results. The model demonstrate strong scalability and effectiveness in detecting zero-day-like attacks, addressing both known and emerging threats in complex IoT environments. Collectively, these studies reflect a growing trend toward enhancing traditional signature based systems through hybridization, optimization, and domain-specific customization.

To conclude, signature based intrusion detection systems still play a vital role in securing connected environments, especially when it comes to identifying well-known attack patterns with high accuracy and minimal false positives. However, their reliance on predefined rules and static signature databases makes them increasingly inadequate against modern, sophisticated, and evolving cyber threats particularly in dynamic and heterogeneous networked systems. As these environments continue to grow in complexity and attackers adopt more adaptive strategies, it becomes evident that traditional signature based methods alone are no longer sufficient. In response, researchers have explored hybrid approaches that integrate signature based detection with anomaly detection, machine learning, or even blockchain technologies to enhance flexibility, detection coverage, and resilience.

To provide a clearer picture of these advancements, Table 8 presents a comprehensive comparative summary of signature based IDS techniques in the literature, outlining their core methodologies, evaluation datasets, performance metrics, and reported limitations. In addition, the table incorporates critical insights that highlight the IoT-specific relevance, applicability, and gaps of each approach. This enriched overview offers not only a synthesis of prior contributions but also a nuanced perspective that clarifies their strengths, weaknesses, and deployment potential in diverse IoT security contexts.

Table 7
Review articles on IDS for IoT.

Year	Authors	Primary focus/scope	Methodological contribution	Identified gaps/future directions	Noted limitations
2024	Roy et al. [143]	Review of “Green” energy-efficient IDS solutions for IoT.	Categorized techniques based on strategies like lightweight anomaly detection and adaptive sampling.	Proposed a framework for sustainable IDS implementations.	Does not extensively explore hybrid IDS architectures and lacks experimental validation of energy-saving methods.
2024	Wu et al. [127]	Extensive review of DL based IDS, including advanced models like Transformers (BERT, GPT).	Categorized a wide range of DL techniques and discussed their advantages for intrusion detection.	Identified key challenges in dataset limitations, feature selection complexity, and model generalization issues.	Primarily focused on DL models; less coverage of other IDS paradigms or deployment architectures.
2024	Halvorsen et al. [145]	Systematic mapping study on the application of Generative Machine Learning Models (GMLMs) in IDS.	Categorized GMLM use cases into penetration testing, dataset augmentation, and standalone IDS.	Highlighted challenges in training stability and the need for standardized evaluation metrics for synthetic data.	Highly specialized on GMLMs; does not integrate these findings into a broader IDS framework.
2024	Al-Haija and Droos [144]	Comprehensive survey on deep learning based IDS in IoT.	Categorized existing IDS methodologies and highlighted the advantages of DL for detecting complex attack patterns.	Suggested future research directions in enhancing real-time IDS performance and developing more efficient learning models.	Noted challenges such as data scarcity, computational overhead, and adversarial robustness.
2024	Kikissagbe and Adda [146]	Comprehensive review of ML based IDS techniques for IoT (supervised, unsupervised, and DL).	Provided a comparative analysis of ML based IDS approaches and their practical applications.	Emphasized the need for adaptable IDS models to handle the heterogeneity and scalability of IoT environments.	Highlights the challenges posed by heterogeneity and scalability as a key limitation of current systems.
2024	Adhikari et al. [132]	Comprehensive review of anomaly detection in IoT.	Categorized techniques into statistical, ML, DL, and hybrid methods.	Identified research gaps and suggested future directions for scalable and adaptive IDS solutions.	Stressed the need for more robust datasets and evaluation frameworks to enhance IDS reliability.
2025	Dritsas and Trigka [147]	Survey on cloud computing applications in IIoT security.	Explored AI-driven intrusion detection, predictive analytics, and anomaly detection within IIoT frameworks.	Explored future research directions such as 5G, serverless computing, and federated learning.	Focuses on the role of cloud integration rather than a broad IDS methodology comparison.
2025	Walling and Lodh [148]	A comprehensive review of ML and DL based IoT IDS systems.	Compared ML and DL based IDSs with detailed analysis on datasets and model validation strategies.	Highlighted ML's efficiency in constrained environments and DL's ability to detect zero-day attacks.	Noted challenges in scalability, real-time performance, and adaptability to real-world IoT scenarios.
2025	Mohale and Obagbuwa [149]	Systematic review of Explainable AI (XAI) in IDS.	Provided a systematic review of XAI techniques (SHAP, LIME, etc.), highlighting the trade-off between interpretability and accuracy.	Proposed hybrid XAI models for enhanced transparency.	Identified challenges such as computational costs and privacy risks while balancing accuracy with interpretability.
2025	Ahsan et al. [150]	Systematic review of metaheuristic based and machine learning-driven IDS approaches in IoT.	Provided an extensive review of IDSs combining metaheuristic optimization techniques with ML and DL models.	Suggested future research directions, including unexplored metaheuristics for IDS enhancements.	Identified critical challenges in scalability, dataset imbalance, and computational constraints in IoT.
2025	Al-Zoubi et al. [142]	Comprehensive survey on IDS in IoT, focusing on ML and DL.	Presented a structured taxonomy of IDS in IoT, evaluating ML, DL, and hybrid approaches.	Discussed critical challenges such as data privacy, scalability, adversarial attacks, and resource constraints.	Emphasized limitations in dataset standardization and model generalizability.
2025	Rahman et al. [151]	Surveys IDS solutions for IoT networks with a focus on ML and DL techniques, datasets, and evaluation metrics.	Analyzed the strengths and weaknesses of current datasets; thoroughly dissected performance metrics and loss functions.	Identified outstanding challenges and recommended potential research directions.	Focuses on the analysis of existing tools (datasets, metrics) rather than proposing new IDS frameworks.
2025	This work	Comprehensive survey of IDS in IoT including industrial applications with a cross-layer taxonomy, blockchain based approaches, and dataset-centric evaluation.	Provides comparative tables of IDS architectures and evaluation metrics; analyzes blockchain-enabled IDS concepts with practical deployment considerations; and supports the discussion with a dataset-driven case study on RPL based routing attacks using multiple machine learning algorithms.	Highlights open challenges such as post-quantum threats, blockchain integration, energy-efficient IDS, federated learning, and dataset standardization for real-world deployment.	This survey delivers a focused and in-depth analysis of cross-layer IDS approaches, spanning, application-layer, and cloud-integrated systems. Rather than concentrating on device-level implementations or specific deployments, the study prioritizes architectural and methodological depth to enable broader generalization and more robust comparison across IDS solutions.

Table 8

Extended comparative summary of signature based IDS techniques in literature.

Year	Authors	Technique & contribution	Dataset/Evaluation	Performance metrics	Limitations	Insights
2011	Meharouech et al. [156]	NAT-aware IDS with alert correlation and attribution	Simulated SCADA environment	Reduced FPR by 15%; improved multi-stage detection	Tailored to NAT networks; lacks IoT-specific evaluation	Demonstrates cross-layer alert correlation potential; could inspire gateway-level IDS in heterogeneous IoT.
2014	Yuxin et al. [155]	Hash based contextual signature filtering	Real enterprise traffic logs	25% fewer false alarms; faster filtering	Preprocessing overhead noted; context binding requires extra resources	Useful for IoT border routers where role- or context-aware alarm suppression is needed.
2016	Punithan et al. [163]	Game-theoretic dynamic signature configuration (DISC)	Custom hybrid framework	92% accuracy; 20% lower storage	High design and tuning complexity	Introduces adaptive signature activation; promising for memory-constrained IoT but requires lightweight adaptation.
2016	Basant et al. [162]	Game-theory based false alarm minimization using Nash equilibrium	KDD'99 dataset	18.7% false alarm reduction	Evaluation limited to outdated datasets; generalization to IoT uncertain	Provides systematic rule tuning concept; basis for IoT auto-tuning IDS with context-aware rules.
2017	Aldwairi et al. [165]	Parallel Myers matching using MapReduce	Hadoop testbed + Snort rules	4× speedup in signature scanning	High compute and resource usage; unsuitable for edge IoT	Demonstrates benefit of cloud-offloading for IoT traffic inspection.
2019	Mutep et al. [157]	Auto-updating signatures with filtering engine	Real-time DB (unspecified)	Enhanced detection through dynamic updates	Dataset unspecified; scalability not validated	Highlights importance of automated signature lifecycle management in IoT.
2019	Wenjuan et al. [164]	Blockchain based collaborative signature IDS	Simulated IoT network	Improved rule trust and verifiability	Blockchain overhead; latency issues in real-time IoT	Pioneers trustable collaborative IDS updates; relevant for multi-stakeholder IoT deployments.
2021	Otoum et al. [159]	Deep Q-learning hybrid IDS (signature + anomaly)	NSL-KDD	Adaptive rule updating; improved detection	High model complexity; real-time tuning difficult	Illustrates RL-driven adaptive IDS concept; needs lightweight RL variants for IoT gateways.
2021	Muthamil et al. [170]	Dual-layer IDS (ML + Snort) in SDN	NSL-KDD	Acc: 93.31%; F1: 0.938; low FPR	Focused on SDN; IoT integration untested	Valuable for IoT-SDN architectures; dual control/data-plane defense blueprint.
2021	Sajad et al. [160]	Suricata + neural nets + fuzzy logic hybrid	Custom simulated network	Acc: 96.11%; strong exploit detection	Limited real-world dataset diversity	Layered approach shows promise for heterogeneous IoT; highlights need for MLOps-driven sustainability.
2022	Amit et al. [166]	Comparative study of ML vs. signature based IDS	NSL-KDD, UNSW-NB15	ML models superior in precision and recall	Lacks IoT focus; comparative only	Underscores why IoT IDS must hybridize ML with signatures rather than rely on one approach.
2023	Erukala et al. [171]	Hybrid IDS for LLNs (specification + signature)	Simulated 6LoWPAN RPL	Accurate detection of RPL based routing attacks	Validation only in simulation	Strong protocol-aware insights for IoT routing security; emphasizes lightweight cross-layer design.
2024	Umer et al. [161]	Meta-learning hybrid IDS for IoMT	IoMT testbed + NSL-KDD	High accuracy; zero-day detection	Meta-model tuning complexity	Novel in IoMT; introduces adaptability that could extend to broader IoT device heterogeneity.

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3.3. Anomaly based detection for IoT security

Anomaly based detection identifies intrusions by analyzing deviations from established behavioral patterns without relying on prior knowledge of specific attack signatures [173]. A baseline profile is created to represent normal network operations, including metrics such as traffic volume, device activity, and communication patterns. Significant deviations from this baseline are flagged as potential threats, making anomaly detection particularly effective for detecting unknown or zero-day attacks. However, the dynamic nature of IoT environments often results in higher false positive rates, where legitimate variations are misclassified as malicious. To enhance detection accuracy, a variety of techniques including statistical analysis, clustering algorithms, machine learning, and deep learning models are employed.

Statistical methods define normal behavior through mathematical modeling and establish thresholds for acceptable deviations in network activity [174]. Time-Series Analysis, Regression Analysis, and Probability Density Estimation are commonly used approaches, offering computational efficiency suitable for resource-constrained IoT devices. Nevertheless, their reliance on static thresholds limits adaptability to evolving behaviors, increasing the likelihood of false alarms. Machine learning techniques address these challenges by dynamically learning complex patterns in network data [175]. ML based IDS approaches can be categorized into supervised and unsupervised learning. Supervised methods, including Support Vector Machines (SVM), Decision Trees, Random Forests, and Artificial Neural Networks (ANNs) [176,177], are trained on labeled datasets and effectively classify known attack types. However, they require frequent updates with new threat signatures to maintain performance [178]. Unsupervised methods, such as

Table 8 (continued).

2024	Schrötter et al. [158]	DLIDS vs. signature based IDS comparison	Simulated IoT env. with Snort	DLIDS better in dynamic env.; Snort stable in static	No new IDS method proposed	Validates that signature based IDS remains reliable baseline in static IoT.
2024	Thankappan et al. [168]	Lightweight signature IDS for Wi-Fi IoT MitM	AWID3; real Wi-Fi IoT setup	TPR: 90% (short); 84% (long)	Specialized only for MC-MitM attacks	Shows feasibility of ultra-lightweight, attack-specific IDS for IoT.
2024	Saravanan et al. [172]	Blockchain + RNN IDS	Malware Executable dataset	Acc: 99.87%; Recall: 99.92%; Precision: 99.66%; F1: 98.38%	Dataset small; scalability not validated	Combines blockchain trust with RNN accuracy; relevant for IoT firmware/malware detection.
2025	Middlesex et al. [167]	Signature based AMQP IDS	Custom AMQP IoT testbed	High accuracy; low FPR	Limited to AMQP; scalability unclear	Fills gap in IoT application-layer security; it proposes the first AMQP-focused IDS signatures.
2025	Alnasser et al. [169]	Hybrid IDS with joint detection DB (IoT + V2G)	CSE-CIC-IDS2018, CIC-DDoS2019	Acc: 95.6%; Prec: 94.2%; Recall: 93.5%; F1: 93.9%	High computational cost in real-time	Extends IDS to IoT + vehicular grids; cross-domain integration is novel.

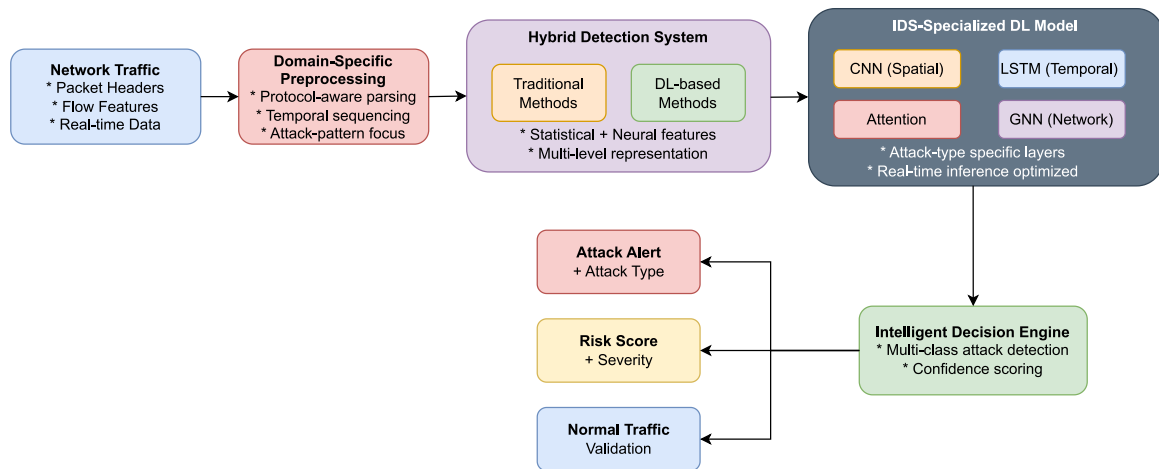


Fig. 8. General workflow of deep learning techniques in intrusion detection systems [185,186].

K-Means clustering and Principal Component Analysis (PCA) [179], detect unknown attacks by discovering anomalies without prior labeling. Additionally, Autoencoders, a type of unsupervised DL model, learn compressed representations of normal traffic and flag deviations as potential intrusions [180].

Deep learning techniques further enhance anomaly detection by leveraging hierarchical feature extraction and learning complex representations of network behaviors [181]. CNNs are widely used for feature extraction from structured network data, identifying spatial patterns indicative of intrusions [177]. RNNs and their variants, such as LSTM networks, are effective in analyzing sequential traffic data and capturing long-term temporal dependencies. LSTM networks specifically address the vanishing gradient problem, which occurs in traditional RNNs when gradients diminish during backpropagation through many layers, hindering the learning of long-range dependencies [182, 183]. Moreover, GANs have been employed to generate synthetic attack traffic that mimics real-world attacks. This process enriches training datasets, helping models generalize better and become more resilient to adversarial threats, where attackers attempt to deceive IDS by crafting data that resembles normal traffic [180,184]. By augmenting datasets with such synthetic examples, IDS models can improve their detection accuracy for both known and previously unseen attacks. Fig. 8 illustrates a deep learning-based IDS workflow for IoT environments. The pipeline includes domain-specific preprocessing followed by a hybrid detection system that combines traditional statistical methods with deep learning techniques. The IDS-specialized DL model incorporates components such as CNNs for spatial features, LSTMs for temporal patterns, GNNs for topological structures, and attention mechanisms for context-aware representation.

The integration of ML and DL into IDS solutions significantly improves detection accuracy, reduces false positives, and enhances adaptability to evolving attack patterns. Furthermore, emerging decentralized learning paradigms such as Federated Learning (FL) have been explored for IDS training without centralizing sensitive IoT data [187]. FL aggregates locally trained models from multiple IoT devices, preserving privacy while enhancing detection capability. However, challenges such as communication overhead, model synchronization, and susceptibility to adversarial attacks remain open research issues [188]. Recent research highlights the potential of hybrid IDS models combining ML and DL approaches [189]. For example, Arafah et al. proposed AE-WGAN, integrating denoising autoencoders with GANs to improve detection of rare and unseen attacks [190]. Gouda et al. introduce a two-layer ML classification framework optimized through feature selection for improved performance [191]. Abdelmoumin et al. compared AML-IDS and DL-IDS models, emphasizing the superiority of deep learning approaches [192]. Other notable contributions include Sattarpour et al.'s BERT based lightweight IDS [193], and Kamal and Mashaly's Transformer-CNN model addressing class imbalance challenges [194]. Lightweight frameworks such as Passban IDS by Eskandari et al. [195], AutoML-optimized IDS for IIoT by Sezgin and Boyacı [196], and fog computing based decentralized IDS by Alatawi et al. [197] have further expanded practical applicability for resource-constrained IoT environments. Blockchain integration into IDS, as explored by Nouman et al. [198], and ensemble learning techniques such as those proposed by Hossain and Islam [199], demonstrate promising directions for future research.

Table 9 provides a comprehensive comparative summary of anomaly based IDS techniques in IoT environments, detailing their

Table 9

Detailed summary of anomaly based IDS and machine learning techniques in literature.

Year	Authors	Technique & contribution	Dataset/Evaluation	Performance metrics	Limitations	Insights
2015	Vidal et al. [200]	Alert correlation for anomaly detection	Not specified	Reduced false positives; improved prioritization	Limited evaluation setup; lacks IoT datasets	Early attempt at anomaly correlation; relevant to IoT log/event fusion but needs protocol-aware design.
2018	Naseer et al. [201]	CNN + LSTM based anomaly detection	NSL-KDD	Improved feature learning; reduced FPR	Uses outdated dataset; no IoT focus	Shows DL's potential; motivates IoT-specific datasets with temporal-spatial patterns.
2019	Tama et al. [202]	Two-stage ensemble classifier IDS (TSE-IDS)	NSL-KDD, UNSW-NB15	Acc ~86%–91%	Limited interpretability; high training cost	Ensemble improves detection robustness; useful for IoT if interpretability + energy efficiency added.
2020	Eskandari et al. [195]	Passban: one-class lightweight IDS	Edge-simulated testbed	Low memory; real-time detection	Evaluation on small-scale synthetic traffic only	Strong for constrained IoT nodes; needs validation with heterogeneous IoT protocols.
2020	Alamiedy et al. [203]	GWO-optimized feature selection with ensemble	NSL-KDD	DoS: 93.6%; Probe: 91.0%	Focuses only on few attack types	Optimization approach applicable to IoT if expanded to full attack surface.
2020	Liu et al. [204]	ML based IDS for Contiki-NG OS	Simulated IoT setup	High accuracy; lightweight	Limited to one OS	Shows OS-specific IDS design; could inspire portable cross-OS models for IoT.
2021	Zachos et al. [205]	RF based lightweight IDS for IoMT	IoMT traffic traces	Low overhead; high TPR	Restricted to healthcare scenarios	Pioneers IoMT-specific IDS; suggests modular IDS for other vertical IoT domains.
2022	Abdelmoumin et al. [192]	AML vs. DL comparative study	UNSW-NB15	DL outperforms AML in accuracy/FPR	No new IDS proposed	Provides baseline; underlines necessity of IoT-centric hybrid benchmarks.
2022	Alatawi and Aljuhani [197]	Fog based anomaly detection (PCA+MRMR)	X-IIoTID	Acc 99.9%; FPR 0.1%	Assumes reliable fog nodes	Valuable for fog/edge IoT; resilience against unreliable fog infrastructure is still open.
2023	Narengbam and Dey [206]	HHO-optimized ANN	AWID dataset	Acc 96.7%; Prec 93.5%	Only compared HHO variants	Shows swarm-optimization in IoT Wi-Fi; broader metaheuristic comparisons needed.
2023	Lara et al. [207]	Modular IDS for smart homes	Real smart home testbed	Low FPR; adaptable	Domain-specific (home only)	Demonstrates modular IDS design; good template for domain-tuned IoT solutions.
2023	Sharma et al. [208]	GAN-augmented DL for rare attacks	UNSW-NB15	DNN: 84%; GAN-DNN: 91%	GAN complexity; training overhead	Tackles class imbalance with GANs; relevant for IoT's long-tail rare threats.
2023	Sezgin and Boyacı [196]	AutoML-driven IDS tuning	IoT traffic testbed	High efficiency; fast tuning	Lacks feature interpretability	Shows promise for fast-deployment IoT IDS; transparency needed for trust.
2023	Nouman et al. [198]	Blockchain + ML hybrid for WSNs	Custom WSN testbed	Acc 98.3%; Prec 97.9%	Blockchain adds compute cost	Strong for collaborative WSN IDS; optimization for ultra-low-power nodes required.
2023	Hossain et al. [199]	Cross-dataset ensemble IDS	NSL-KDD, CICIDS2018, WSN-DS	Acc >99% across datasets	Real-time feasibility unproven	Valuable demonstration of generalizability; IoT real-time trials needed.
2024	Sattarpour et al. [193]	BERT based EBIDS (network/app layer)	CIC-DoS2017, Edge-IIoT	High accuracy; low overhead	Needs broader benchmarking	Introduces NLP-style embeddings into IoT IDS; promising for application traffic.
2024	Kamal and Mashaly [194]	Transformer-CNN with imbalance handling	UNSW-NB15	High multiclass detection	Heavy training for imbalanced data	State-of-art DL; effective for IoT datasets if optimized for edge.

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Table 9 (continued).

2024	Seniaray and Jindal [209]	DMI-GA feature selection + ML	IoTID20	Acc 99.9%; 34% faster	GA sensitive to parameters	IoTID20 focus; relevant for low-resource IoT if robustness improved.
2024	Al-Fuhaidi et al. [210]	ML-IDS with SMOTE + RFE	Simulated WSN dataset	Acc 98.9%; Prec 98.4%; F1 97.9%	Simulation-only; no real deployment	Shows IoT-WSN potential; needs testbed validation.
2024	Nguyen et al. [211]	DNCAE/DCHAE clustering	Public datasets + X-means	Improved zero-day detection	Complexity of clustering	Valuable for IoT zero-day defense; requires scaling for constrained devices.
2024	Gouda et al. [191]	Two-layer ML IDS with FS	NF-UQ-NIDS-v2	RF: 99.1%; LSTM: 98.9%	High accuracy; low interpretability	Strong benchmarks; interpretability key for IoT deployment trust.
2025	Arafah et al. [190]	AE-WGAN rare attack detector	NSL-KDD, CICIDS2017	Good for minority attacks	GAN instability risk	Highlights IoT-tailored rare attack detection; adversarial robustness required.
2025	Popoola et al. [212]	Multi-Stage Deep Learning (MSDL) for IIoT	X-IIoTID, WUSTL-IIoT	P/R/F1 >99.7%	High runtime complexity	High accuracy for IIoT; real-time deployment remains challenge.

methodologies, datasets, and performance metrics. In addition to reported limitations, the table incorporates critical insights that highlight IoT-specific applicability, strengths, and open challenges, offering a nuanced perspective on their practical effectiveness and deployment trade-offs.

In summary, anomaly based detection in IoT IDS has evolved significantly with the integration of machine learning and deep learning techniques. These advancements have enabled more accurate, scalable, and adaptive intrusion detection, although challenges remain in terms of computational efficiency, data scarcity, and resilience against adversarial attacks. Future research should prioritize lightweight model design, real-time adaptability, and privacy-preserving distributed learning to address these open issues effectively.

3.4. Specification based detection techniques

Specification based detection enforces predefined behavioral rules for system operations, ensuring that IoT devices function within established parameters. Unlike signature based detection, which relies on known attack patterns, or anomaly based detection, which dynamically learns behavioral deviations, this approach operates on expert-crafted specifications. Any deviation from these predefined rules is considered a potential security threat. This method is particularly effective for IoT environments, where devices often exhibit predictable operational patterns. For instance, an IoT thermostat is expected to communicate only with authorized servers at fixed intervals. Any communication outside this defined behavior, such as unexpected remote access or anomalous data transmissions, can be flagged as an intrusion. The primary advantage of specification based detection lies in its precision, resulting in lower false positive rates compared to anomaly based techniques. However, the effectiveness of this method depends heavily on the accuracy and completeness of predefined specifications. Developing these specifications requires expert domain knowledge and continuous updates to accommodate evolving IoT applications and emerging attack strategies. Additionally, the rigid nature of rule based enforcement may limit adaptability to novel or unforeseen threats, making it less effective in highly dynamic IoT ecosystems.

Several recent studies have demonstrated the effectiveness of specification based detection in IoT security. Sekar et al. introduce one of the earliest hybrid models that combine specification based detection with anomaly detection using protocol state machines, achieving both high accuracy and low false positive rates on benchmark datasets [213]. Hussein and Zulkernine propose an early specification based IDS framework for component based systems using the UMLIntr profile. Their approach enables automatic intrusion signature generation from UML models and was validated with DARPA datasets [214]. Berthier and Sanders introduce a specification based intrusion detection

sensor designed for Advanced Metering Infrastructures (AMI), a critical component of smart grid ecosystems. Their system enforces compliance with predefined behavioral constraints derived from the C12.22 communication protocol, operating across the network, transport, and application layers to monitor traffic between smart meters and access points. The authors formally verify the correctness of the specified constraints and implement a real-time detection prototype capable of identifying policy violations in live AMI environments. This approach enhances situational awareness in power distribution networks and provides a scalable, field-deployable IDS solution tailored to the operational characteristics of smart meters [215]. Le et al. propose a specification based IDS to detect topology attacks in RPL, which is widely used in IoT networks [217]. Their system constructs an Extended Finite State Machine (EFSM) from network simulation traces to model legitimate RPL operations, utilizing cluster based monitoring and sequence tracking in control messages (DIO/DIS) to enhance detection accuracy while maintaining a low overhead (6.3%). Similarly, Yun et al. introduce a behavior-rule specification based IDS tailored for safety-critical embedded devices in smart homes [220]. Their approach defines behavior rules for devices such as fire, gas leak, and humidity sensors, modeling them through UML state machines. The system demonstrated high detection accuracy even under noisy conditions up to 50%, outperforming conventional classifiers like SVM and Decision Tree in both detection performance and resource efficiency, including lower CPU usage (1.47%) and memory consumption (49 kB).

Babu and Reddy introduce Specification Heuristics Based IDS (SH-IDS), which classifies IoT network transactions using n-gram sequential pattern analysis and rule based thresholds [219]. Their system achieved 91.7% accuracy on the UNSW-NB15 dataset, outperforming contemporary IDS models while maintaining a low false positive rate and efficient memory and energy utilization, making it suitable for resource constrained IoT networks. Sharma et al. propose BRIoT, a behavior-rule specification based IDS leveraging hierarchical context-aware Petri nets and formal verification for detecting misbehavior in IoT-embedded cyber-physical systems [218]. Their system demonstrated high detection accuracy with minimal overhead, proving effective even in real-time UAV environments. Mitchell and Chen develop a behavior-rule specification based IDS (BSID) for medical cyber-physical systems (MCPS), defining expected behaviors for medical devices and monitoring compliance via state machines [216]. Their system effectively detected sophisticated and hidden attacks, maintaining patient safety and minimizing false positives compared to anomaly based IDS models.

In summary, specification based detection in IoT IDS offers a highly precise and interpretable approach by enforcing predefined behavioral rules tailored to device operations. This method is particularly effective in environments where IoT behaviors are deterministic and predictable, enabling low false positive rates and strong compliance enforcement. As

Table 10

Detailed summary of specification based IDS techniques in literature.

Year	Authors	Technique & contribution	Evaluation dataset/domain	Performance metrics	Limitations	Insights
2002	Sekar et al. [213]	Protocol state machine-based hybrid IDS (specification + anomaly)	DARPA 1999 dataset	Low false alarm rate; high detection accuracy	Based on outdated DARPA'99 traces; modern IoT protocols not addressed	Early hybridization of specification + anomaly detection; inspires lightweight protocol state machines for IoT.
2007	Hussein & Zulkernine [214]	UMLIntr: UML based intrusion modeling with automatic signature generation	DARPA 1999 (simulated scenarios)	All modeled intrusions detected in prototype	Domain-limited to component based software; lacks real-world deployment	Pioneering automatic model-to-signature approach; relevant for embedded IoT software design.
2011	Berthier & Sanders [215]	Specification based IDS for AMI using C12.22 formal constraints	Smart meter networks; AMI traffic traces	Multi-layer monitoring; validated detection logic	Tailored to AMI; portability to broader IoT protocols unclear	Strong case for formal verification in IoT-specific protocols; early example of smart grid IDS.
2015	Mitchell & Chen [216]	Behavior-rule IDS for medical CPS (MCPS)	Simulated medical CPS	High detection accuracy; low false positive rate	Manual rule definition needed; limited scalability	Highlights rule-to-state-machine transformation; applicable to healthcare IoT device monitoring.
2016	Le et al. [217]	Specification based IDS for RPL topology attacks in 6LoWPAN	Simulated 6LoWPAN with RPL	Lightweight; accurate against topology manipulation	Evaluation limited to simulation; only RPL manipulation considered	Demonstrates protocol-aware specification rules; directly relevant to IoT routing security.
2019	Sharma et al. [218]	BRIoT: Behavior-rule IDS with formal model checking for UAV CPS	UAV based IoT environment (simulated)	High detection rate; low false positives	Domain-limited to UAV systems; generalizability not shown	Shows feasibility of formal verification for embedded IoT CPS; addresses zero-day threats with lightweight rules.
2020	Babu & Reddy [219]	SH-IDS: Heuristic-driven specification IDS for LLNs	Simulated 6LoWPAN IoT networks	Accuracy ~91.7%; low FPR; low resource usage	Tested only in small-scale LLNs; lacks real deployment	Integrates heuristic rule derivation with spec-based IDS; valuable for resource-constrained IoT devices.
2021	Yun et al. [220]	Behavior-rule IDS for smart homes with UPPAAL verification	Real smart home testbed	Accuracy robust under 50% noise; CPU 1.47%, RAM 49 KB	Scalability to multi-user/larger deployments untested	Demonstrates practical smart-home IDS; strong fit for consumer IoT with formal verification support.

reflected in recent studies, specification based techniques have demonstrated success across domains such as smart grids, UAVs, medical systems, and smart homes through formal modeling, behavior rule definitions, and protocol-specific monitoring. However, the need for expert-crafted specifications, limited scalability, and difficulty adapting to evolving or unforeseen attacks remain key limitations. To address these challenges, future research should prioritize the automation of rule generation, hybridization with anomaly or learning based methods, and the development of scalable, lightweight frameworks suitable for dynamic and heterogeneous IoT ecosystems.

Table 10 provides a comprehensive comparative summary of specification based intrusion detection techniques, detailing their methodologies, application domains, and evaluation environments. Beyond reporting performance metrics and limitations, the table integrates critical insights that distill each study's IoT-specific relevance, scalability, and practical deployment challenges, thereby delivering a clearer and more rigorous perspective on their effectiveness in securing IoT and cyber-physical systems.

3.5. Hybrid intrusion detection methods

Hybrid intrusion detection systems integrate multiple detection techniques to enhance security in IoT networks by addressing the limitations of individual approaches [221]. Given the diverse attack surface of IoT environments, hybrid methods combine signature, anomaly, and specification based detection strategies to create a more robust and adaptable defense mechanism [222]. By leveraging the strengths of

these techniques, hybrid IDS improve detection accuracy, minimize false positives, and enhance adaptability to emerging cyber threats.

While signature based detection provides high accuracy in identifying known attacks, it struggles against novel threats. In contrast, anomaly based detection effectively detects unknown attack patterns but tends to produce false positives due to the dynamic nature of IoT networks. Specification based detection ensures compliance with predefined operational behaviors but may fail against sophisticated attackers operating within the expected parameters. To bridge these gaps, hybrid systems often use a layered approach, where signature based detection filters out known threats, anomaly based detection identifies behavioral deviations, and specification based detection enforces operational constraints. This combination ensures a well-rounded security framework suitable for the unique constraints of IoT ecosystems.

Despite their advantages, hybrid IDS implementations introduce computational overhead, requiring optimization strategies for resource constrained IoT devices [223]. Researchers are increasingly exploring lightweight hybrid IDS models leveraging edge computing, federated learning, and distributed detection architectures to mitigate performance bottlenecks. Several notable hybrid IDS models have been proposed in the literature. Napiiah et al. introduce the Compression Header Analyzer IDS (CHA-IDS), a hybrid system that combines anomaly and signature based techniques to detect malicious activities in the 6LoWPAN protocol [224]. By analyzing compression headers, CHA-IDS effectively identified individual and combined attacks, such as hello flood, sinkhole, and wormhole. Utilizing machine learning methods for feature selection and classification, CHA-IDS achieved 99% detection

accuracy, demonstrating resource efficiency suitable for constrained IoT devices. Sawafi et al. [225] propose a hybrid deep learning based intrusion detection system specifically designed for RPL based IoT networks in 2023. Their model integrates a supervised Domain-Adversarial Neural Network (DANN) with a semi-supervised Deep Autoencoder (DAE), enabling the detection of known and unknown threats. The system is evaluated using IoTR-DS, a newly developed dataset containing representative IoT attacks such as DIS flood, Sequence manipulation, and Wormhole attacks. When tested on pre-trained (known) attack traffic, the model achieved 98% accuracy and 92% multi-class F1-score. For untrained (unknown) attack scenarios, the model achieved strong generalization, achieving 95% accuracy and 87% F1-score. Khan et al. propose a hybrid deep learning model based on RNN-GRU that encompasses the physical, network, and application layers, achieving successful results with accuracies up to 99.21% on the ToN-IoT dataset [226]. While this model offers a robust solution for classifying IoT attacks, it has not been sufficiently explored on resource-constrained devices.

Similarly, Kim et al. develop a hierarchical hybrid intrusion detection model that integrates anomaly based and misuse based (signature based) detection methodologies [227]. Their system employs a decision tree based misuse detection module followed by a one-class SVM for anomaly detection. Evaluations on the NSL-KDD dataset demonstrate improved detection rates for both known and unknown attacks, while reducing false positives and achieving significant reductions in training and testing times, thus making it suitable for real-time IoT security. Building on this line of hybrid architectures, Ramadan and Yadav propose a two-stage hybrid IDS architecture that integrates a LCNN with a GRNN, enhanced through an ESFL based feature selection mechanism [228]. Their system is evaluated on the NSL-KDD dataset, specifically on the KDDTest+ and KDDTest-21 subsets, where it achieved 97.89% and 92.56% accuracy, respectively. The hybrid model demonstrated strong detection capabilities across both known and complex attacks (DoS, U2R, R2L), while maintaining low false positive rates. The layered design, while effective, may introduce latency and computational cost in highly resource constrained environments. In a parallel effort focused on ensemble learning, Khraisat et al. propose a novel ensemble based hybrid intrusion detection system that integrates anomaly and signature based techniques through machine learning algorithms such as C5.0, NB, SVM, CART, KNN, and DT classifiers [229]. Their system emphasizes combining the strengths of both detection types to improve accuracy and reduce false alarms. Experimental evaluation on the BoT-IoT dataset demonstrated strong results, with a detection rate of 99.97%. However, the study heavily relies on a single dataset, and the generalizability of the results to other IoT contexts or real-world deployments remains unvalidated.

To summarize, hybrid intrusion detection systems offer a promising pathway to address the evolving security challenges of IoT environments by combining the strengths of multiple detection paradigms. Recent research demonstrates how hybrid models leverage machine learning, deep learning, cryptographic, and ensemble techniques to improve detection accuracy across different layers of the IoT protocol stack. While these systems achieve notable results, challenges such as computational complexity, latency, and limited validation in real-world deployments remain significant particularly for resource constrained or highly dynamic IoT scenarios. To provide a clearer and more rigorous perspective on these developments, Table 11 presents a comprehensive overview of hybrid IDS approaches, detailing their detection strategies, application domains, performance outcomes, reported limitations, and IoT-specific insights that illuminate their practical effectiveness and deployment trade-offs.

To provide a clearer synthesis of the relative strengths and weaknesses of each IDS method discussed, Table 12 summarizes their comparative analysis.

3.5.1. Signature based and anomaly based detection

The combination of signature and anomaly based detection is widely adopted in hybrid IDS designs [232]. Signature based detection enables rapid identification of known attacks through predefined patterns with minimal computational overhead. However, it is inherently limited to previously identified threats and requires continual updates to the signature database. Anomaly based detection complements this by identifying deviations from established normal behavior, allowing for the detection of previously unseen or zero-day attacks [233]. For example, in an IoT network, signature based detection may effectively block well-known DDoS patterns, while anomaly detection monitors traffic for behaviors that do not align with established norms, potentially indicating emerging threats. Despite these strengths, maintaining an accurate normal behavior profile and controlling false positives in anomaly detection remain key challenges. Beyond these traditional methods, machine learning techniques have been increasingly integrated to enhance IDS effectiveness. ML models such as deep neural networks, clustering algorithms, and ensemble methods excel at uncovering complex patterns within traffic data, thereby improving the detection of sophisticated and evolving threats that may elude both signature and anomaly based methods. Furthermore, machine learning assists in dynamically refining signature databases by extracting new threat signatures from observed attack patterns [233]. In summary, while signature and anomaly based detection provide a foundational dual-layer approach targeting known and unknown threats respectively the incorporation of machine learning introduces an adaptive and intelligent mechanism. This synergy enhances both the detection capability and the resilience of IDS, especially in dynamic and resource constrained environments like IoT networks.

3.5.2. Specification and anomaly based detection

Integrating specification and anomaly based detection enables the development of hybrid IDS that combine the strictness of predefined operational rules with the flexibility of detecting unknown threats. Specification based detection operates by enforcing formally defined behavior models of systems or network protocols, thereby minimizing false positives. However, it can be limited by the manual effort required to define comprehensive specifications. Anomaly based detection complements this by identifying deviations from normal behavior, capturing unknown or sophisticated attacks that may not violate explicit specifications but still exhibit abnormal characteristics. While this enhances adaptability, anomaly based methods are often prone to higher false positive rates. To mitigate the limitations of each method, hybrid models have been proposed that integrate both techniques. For example, Bostani and Sheikhan [231] propose a hybrid IDS for IoT that combines specification based detection at router nodes with anomaly detection using an unsupervised Optimum-Path Forest (OPF) algorithm based on the MapReduce framework. This design facilitates the detection of routing attacks such as sinkhole and selective forwarding in IoT networks by merging localized rule enforcement with global anomaly analysis. Similarly, Stakhanova et al. [230] present a framework that fuses specification and anomaly based detection by automatically generating specifications through machine learning classifications. Their approach employs the EXtended ACTION graph to store and update variable-length behavioral patterns, allowing the IDS to adaptively refine its detection capabilities over time. These approaches demonstrate that combining specification based and anomaly based detection can provide a balanced strategy that offers both precision in known behavior enforcement and adaptability in identifying emerging threats, particularly in dynamic and resource constrained environments such as IoT.

Table 11
Detailed summary of hybrid IDS techniques in literature.

Year	Authors	Technique & contribution	Evaluation dataset/domain	Performance metrics	Limitations	Insights
2010	Stakhanova et al. [230]	Hybrid IDS combining specification and anomaly detection using Extended ACTion Graph (Exact)	Simulated system call traces	Detected known and novel threats; low false positive rate	Validated only in simulation; no IoT-specific protocols tested	Early hybridization concept; inspires IoT-oriented hybrids but needs protocol-aware adaptation.
2014	Kim et al. [227]	Hierarchical hybrid IDS (Decision Tree misuse + One-class SVM anomaly)	NSL-KDD dataset	High detection rate for known/unknown attacks; low FPR; reduced training/testing time	Complexity of tuning multiple models; may limit real-time use	Strong hierarchical design; shows potential for IoT gateways if optimized for constrained hardware.
2017	Bostani & Sheikhan [231]	Optimum-Path Forest + MapReduce hybrid for routing attack detection	Simulated IoT routing environment	Accurate sinkhole and selective forwarding detection	No public IoT dataset used; scalability assumptions only	Highlights graph based anomaly + spec integration; useful for RPL/LLN attack scenarios.
2018	Napiah et al. [224]	CHA-IDS: Hybrid IDS for 6LoWPAN using compression header analysis	6LoWPAN testbed	Accuracy ~99%; low overhead	Focused on 6LoWPAN only; limited attack diversity	Demonstrates lightweight hybrid tailored to constrained IoT stacks.
2019	Kalavadekar et al. [232]	Hybrid IDS (Genetic Algorithm for signature + Association Rule Mining for anomaly)	KDD'99, NSL-KDD, Kyoto 2006+, UGR16	Accuracy up to 99.9%	High training cost; lacks real-time validation	Shows multi-dataset robustness; highlights training-cost trade-off critical for IoT IDS.
2019	Khraisat et al. [229]	Ensemble hybrid (C5.0, SVM, CART, KNN, NB, DT) on BoT-IoT	BoT-IoT dataset	Accuracy 99.97%	Evaluated only on one dataset; not tested in real IoT networks	Proves ensemble hybrids excel on IoT-specific datasets; requires generalization across domains.
2020	Ramadan & Yadav [228]	Hybrid IDS with LCNN + GRNN; ESFL based feature selection	NSL-KDD (KDDTest+, KDDTest-21)	Acc 97.9%, FPR 0.72% (KDDTest+)	Benchmark datasets outdated; no IoT-specific validation	Novel lightweight DL hybrid; motivates applying optimized DL to IoT traffic.
2023	Sawafi et al. [225]	Hybrid IDS evaluated on IoTR-DS (RPL routing attacks)	IoTR-DS (DIS, Rank, Wormhole)	Acc 98% (pre-trained); F1-score 92%	Limited to RPL based IoT; dataset scope narrow	First hybrid IDS tested on IoT-RPL dataset; relevant to LLN routing security.
2023	Khan et al. [226]	RNN-GRU-based hybrid IDS across IoT layers	ToN-IoT dataset (Telemetry, NetFlow, Windows logs)	Acc 99.2%; Prec 98.9%; Recall 98.9%; F1 98.9%	Not tested on constrained real IoT deployments	Multi-source validation shows promise; high accuracy across IoT layers; resource costs remain concern.
2025	Nandanwar et al. [223]	Hybrid Blockchain-ML IDS (ECC, DSA, SHA-512 + XGBoost)	Edge-IIoT dataset	Acc 98.1%; F1 97.9%; FPR 2.2%; Blockchain latency 0.34 s	Dataset lab-based; blockchain overhead may hinder real-time use	Integrates cryptography + ML; scalable trust-aware IDS for heterogeneous IoT.
2025	Ahmed et al. [233]	Adaptive hybrid IDS (signature + ML/DL with fuzzy clustering)	Real-world datasets (unspecified)	Reported high precision; scalable	Dataset not disclosed; reproducibility unclear	Highlights adaptability and scalability; but transparency limits academic/industrial uptake.

3.6. Comprehensive datasets for intrusion detection in IoT networks

IDS rely heavily on comprehensive, diverse, and representative datasets to ensure effective training, evaluation, and benchmarking. These datasets enable the development of IDS by providing labeled and unlabeled traffic data, covering a wide range of normal behaviors and attack scenarios. Publicly available datasets are especially critical in standardizing comparative evaluations and ensuring reproducibility in IDS research. Over the past two decades, a variety of datasets have been proposed to reflect evolving network architectures and attack landscapes, including IoT specific threats [234]. In this work, we consider a dataset to be IoT-relevant if it either includes traffic from IoT devices or has been used extensively in IoT IDS studies, particularly in contexts involving lightweight protocols, resource constraints, or edge deployment scenarios. Table 13 provides a detailed overview of notable

publicly available datasets relevant to IDS in IoT environments. Each dataset is summarized in terms of its content, scale, covered attack types, and usage in the literature.

While these datasets provide an essential foundation for IDS research, several limitations and methodological issues deserve close attention. Firstly, many classical datasets such as KDD-99 and NSL-KDD, although still widely used, were originally designed for legacy IT environments and do not reflect the traffic patterns, protocols, or resource constraints characteristic of modern IoT systems. They also often lack attack diversity and include outdated threat vectors that do not represent current adversarial tactics. Modern datasets such as Edge-IIoTset, BoT-IoT, IoT-23, TON_IoT, CICIoT2023 and IDSIoT2024 attempt to address these gaps by incorporating recent attack types, real IoT devices, or complex traffic patterns. However, a closer inspection reveals that many of these are still generated in synthetic or

Table 12
Comparative analysis of IDS methods in IoT.

IDS method	Strengths	Weaknesses
Signature based IDS	High accuracy for known attacks; Low false positives	Ineffective against unknown or zero-day attacks; Requires continuous signature updates
Anomaly based IDS	Detects unknown and novel attacks; Adaptive to evolving threats	High false positive rates; Requires large and well-labeled datasets for training
Specification based IDS	Low false positives due to strict behavior rules; Effective for well-defined systems	Difficult to define comprehensive specifications; Limited flexibility to new behaviors
Hybrid IDS	Combines advantages of multiple methods; Improved detection accuracy and adaptability	Increased computational overhead; More complex implementation

simulated environments, often with scripted traffic, simplified topologies, and limited device diversity. This raises questions about the suitability and generalizability of IDS models trained and evaluated solely on such data. Another common issue is the class imbalance found in many datasets, where benign traffic may significantly outnumber attack instances, or vice versa. Without appropriate preprocessing or sampling techniques, this imbalance can lead to biased training and misleading performance metrics particularly inflated accuracy or precision [371]. Additionally, many datasets do not disclose details regarding labeling methodologies, making it unclear whether annotations are derived from known ground truths, manual tagging, or inferred heuristics. The reproducibility of dataset based evaluations also remains a challenge. Studies using the same dataset may report widely varying results due to differences in feature selection, data partitioning, or normalization techniques. Without standardized protocols for dataset usage, benchmarking across studies remains inconsistent and potentially unreliable [372].

From a practical deployment perspective, most datasets also fail to capture real-time performance constraints such as CPU usage, memory consumption, latency, or energy overhead all of which are critical for deploying IDS in resource constrained IoT domain. Furthermore, modern threats such as adversarial machine learning attacks, poisoning of federated IDS models, and encrypted traffic analysis are largely absent from existing datasets, leaving a blind spot in IDS development. While recent efforts such as CICIOT2023 and IDSIoT2024 offer improvements through longer collection durations, greater device heterogeneity, and real hardware deployment, they still face challenges in terms of scalability, availability, and documentation clarity. Many datasets do not provide full details about network configuration, instrumentation tools, attack injection methods, or how normal behaviors were established all of which are crucial for meaningful replication and evaluation [373].

In summary, although the growing availability of IoT focused IDS datasets has greatly benefited the research community, further progress requires more realistic, transparent, and standardized dataset generation practices [374]. There is also a pressing need for datasets that capture emerging attack types, diverse device behaviors, and dynamic protocol traffic, while remaining accessible and well-documented. Finally, future datasets should prioritize support for cross-dataset generalization evaluations, enabling researchers to test IDS robustness beyond a single controlled environment a critical step toward practical and resilient IDS deployment in heterogeneous and evolving IoT ecosystems.

3.7. Case study

In this case study, we assess the effectiveness of various machine learning algorithms in detecting RPL based routing attacks in an IoT environment using the ROUT-4-2023 dataset [360]. This dataset is created in the Contiki network simulator [375,376] and reflects realistic traffic patterns under both benign and attack conditions.

3.7.1. Dataset description

The ROUT-4-2023 dataset includes traffic data from simulations involving four major RPL-specific attacks: Blackhole, Flooding, DODAG Version Number, and Decreased Rank. Each attack scenario consists of both malicious and normal traffic, enabling binary classification [361, 377]. Table 14 presents the distribution of samples.

A total of 16 traffic-level features are identified, including control message counts, transmission statistics, and timing metrics. Table 15 lists the extracted features.

During preprocessing, the features TIME, SOURCE, and DESTINATION are excluded, as these are node-specific and not generalizable in dynamic networks. The final 13 features are used to train and evaluate the models.

3.7.2. Performance evaluation

A 5-fold cross-validation strategy is used to guarantee a thorough and objective performance evaluation. To enable each instance to perform both functions across various folds, the dataset is randomly divided into training and testing subsets that are mutually exclusive in each iteration. In order to minimize the variance caused by random sampling, the final performance metrics are calculated as the average across all five folds. Four commonly used evaluation metrics that are derived from the confusion matrix are used to evaluate the effectiveness of the classifier:

- **True Positive (TP):** Correctly identified attack instances.
- **True Negative (TN):** Correctly identified normal (non-attack) instances.
- **False Positive (FP):** Normal instances incorrectly classified as attacks.
- **False Negative (FN):** Attack instances incorrectly classified as normal.

Using these values, the following metrics are defined:

Accuracy. represents the proportion of all correct predictions (both positive and negative) over the total number of predictions:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (1)$$

Recall. measures the model's ability to correctly identify true attack instances:

$$\text{Recall} = \frac{TP}{TP + FN} \times 100 \quad (2)$$

Precision. evaluates how many of the instances predicted as attacks are actually attacks:

$$\text{Precision} = \frac{TP}{TP + FP} \times 100 \quad (3)$$

F1 Score. is the harmonic mean of precision and recall, providing a balanced measure:

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \times 100 \quad (4)$$

The evaluation results for eight selected algorithms are shown in Table 16. As the results demonstrate, ensemble based methods (e.g., Bagging, Random Forest) and Decision Tree classifiers achieved near-perfect performance. In contrast, baseline models such as Logistic Regression and Gaussian Naive Bayes performed relatively poorly on the dataset.

Table 13

Publicly available datasets for IDS in IoT environments.

Dataset name	Year	Description	Type of event	Dataset size	Refs.
KDD-99 [235]	1999	Benchmark IDS dataset containing labeled traffic data for a range of simulated attacks and normal connections.	Smurf, Neptune, Back, Teardrop, and others	≈900 MB	[236–240]
NSL-KDD [241]	2009	Enhanced version of KDD-99 addressing redundancy issues and providing a balanced evaluation set.	User to Root, Remote to Local, DoS, Probe	≈100 MB	[242–245]
ISCX-2012 [246]	2012	Labeled network traffic dataset designed for evaluating intrusion detection systems, includes normal and malicious behaviors.	DoS, DDoS, Infiltration, HTTP, Botnet, Brute force SSH	≈85 GB	[176,247–252]
CTU-13 [253]	2013	A botnet traffic dataset containing 13 labeled scenarios (PCAP + bidirectional NetFlows) with real-world malware, background and normal traffic.	Rbot, Virut, Menti, Sogou, Murlo, and NSIS	≈696 GB	[254–259]
UNSW-NB15 [260]	2015	Modern dataset featuring realistic normal and malicious network traffic collected in a hybrid testbed environment.	Generic, Exploits, Fuzzers, Reconnaissance, DoS	≈14.16 MB	[261–269]
WSN-DS [270]	2016	Dataset targeting wireless sensor networks with various attack scenarios relevant to IoT applications.	Grayhole, Blackhole, Flooding, Scheduling	≈26.61 MB	[139,198,199,204, 224,271,272]
CIDDS-001 [273]	2017	Dataset capturing realistic network traffic with both external and internal attacks, aimed at anomaly detection research.	Port Scan, Brute Force, DoS, DDoS, Ping Scan	≈4.4 GB	[274–278]
CICIDS2017 [279]	2017	Comprehensive dataset simulating benign and malicious traffic based on contemporary attack scenarios.	Brute Force FTP, DoS, Botnet, Heartbleed, Port Scan, Web Attack	≈60.1 GB	[280–283]
CIDDS-002 [284]	2018	Extension of CIDDS-001, focusing on updated attack profiles and network configurations.	SYN, ACK, FIN Scans, Ping Scans	≈2.2 GB	[285,286]
DS2OS [287]	2018	Dataset for intrusion detection in IIoT environments using the DS2OS architecture.	DoS, Malicious Control, Scanning, Spying	≈125.3 MB	[288–290]
N-BaIoT [291]	2018	Dataset focusing on botnet attacks on IoT devices such as webcams and baby monitors.	Scan, UDP Flooding, TCP Syn, Junk Traffic	≈8.15 GB	[292–296]
CICIDS2018 [297]	2018	Enhanced version of CICIDS2017, including additional attack types relevant to IoT systems.	SSH Brute Force, Web Attacks, DDoS, Botnet Activity, Infiltration	≈200 GB	[298–302]
BoT-IoT [303]	2019	Large-scale dataset simulating botnet traffic targeting IoT networks in various attack scenarios.	DDoS, Service Scan, Keylogging, Data Exfiltration	≈69.3 GB	[304–306]
CICDDoS2019 [307]	2019	Dataset for DDoS detection including both PCAP and flow based formats. It features recent, high-traffic reflective and exploitation based DDoS attacks.	PortMap, NetBIOS, MSSQL, UDP, UDP-Lag, SYN, NTP, DNS, SNMP, SSDP, TFTP, Web-DDoS	≈13.2 GB	[308–310]
IoT-23 [311]	2020	Dataset containing diverse malware traffic targeting IoT devices, including Mirai and Gafgyt botnets.	DDoS, Heartbeat, File Download, Malware Infection	≈21 GB	[312–317]
TON_IoT [318]	2020	Comprehensive dataset for IoT, IIoT, and cloud intrusion detection scenarios.	Backdoor, DDoS, Injection, MITM, XSS, Password Attacks	≈50 GB	[319–325]
Routing Attack Dataset for RPL based IoT [326]	2020	PCAP and LOG files simulating routing attacks on RPL based IoT networks.	Version Number Attack, Hello Flood, Worst Parent Attack	≈953.7 MB	[327–333]

(continued on next page)

3.7.3. Discussion of results

The results presented in Table 16 clearly demonstrate the superiority of tree based ensemble methods, particularly Bagging, Random Forest, and Decision Tree, in detecting RPL based routing attacks. These models consistently achieve nearly perfect classification performance, with accuracy, precision, recall, and F1 scores all exceeding 99.9%. The high performance of these algorithms can be attributed to their ability

to model complex non-linear decision boundaries and reduce overfitting through techniques like bootstrapping and aggregation. The KNN algorithm also performs exceptionally well, achieving above 99.7% in all metrics. However, its computational cost in prediction time may be a limiting factor in real-time IoT deployments, where resource constraints are critical. Gradient Boosting provides strong results (accuracy of 90.65%), but it under-performs compared to Bagging-based methods.

Table 13 (continued).

IoTID20 [334]	2020	Dataset containing network traffic of IoT devices under multiple attack scenarios. Data collected using smart home devices with traffic captured via monitor mode.	Traffic includes benign and various attacks such as ARP spoofing, SYN, port scanning, OS scanning, and Mirai Botnet behaviors (UDP, ACK, HTTP flooding, Host Bruteforce)	≈295 MB	[335–338]
IoT-CIDDS [339]	2021	Labeled dataset designed for intrusion detection in 6LoWPAN based IoT networks, generated using a real testbed with Contiki OS and Cooja emulator.	Blackhole, selective forwarding, hello flood, UDP flooding, RPL DIS flooding, ICMPv6 flooding attacks	≈50 MB	[340,341]
AWID3 [342]	2021	PCAP files capturing extensive 802.11 enterprise network attacks with PMF enabled, along with multilayer and 802.3 network assaults	Deauthentication, Disassociation, (Re)Association, Rogue AP, Krack, Kr00k, Botnet	≈33.9 GB	[343–346]
UQ-IOT-IDS-2021	2021	Realistic IoT dataset collected in a university network testbed using Wireshark. Includes benign traffic from smart TVs, cameras, phones, Raspberry Pis, and malicious traffic from 9 attack types.	Host Discovery, Port Scanning, Service Detection, ARP Spoofing, Telnet Brute-force, SYN/UDP/HTTP/ACK flooding	≈70.8 GB	[347–349]
Edge-IIoTset [350]	2022	Comprehensive IoT/IoT dataset for centralized and federated learning IDS, generated via a diverse testbed with over 10 device types and multiple protocols.	DoS/DDoS, Info Gathering, MITM, Injection, Malware	≈1.4 GB	[351–355]
X-IIoTID [356]	2022	Device and connectivity dataset designed for IIoT systems covering diverse attack scenarios, protocols, and multiview features.	Botnet, MITM, Spoofing, Data Exfiltration, Scanning, Malware	≈355 MB	[357–359]
ROUT-4-2023 [360]	2023	Recent dataset capturing RPL routing attacks simulated in Cooja for ML based IDS development.	Blackhole, Flooding, Decreased Rank, DODAG Version Attack	≈311 MB	[24,361–363]
CICIoT2023 [364]	2023	Real-time IoT dataset with 33 attack types in 7 categories, collected using a topology of 105 real IoT devices.	DDoS, DoS, Recon, Web-based, Brute Force, Spoofing, Mirai	≈13.32 GB	[308,336,365, 366]
IDSIoT2024 [367]	2024	Real-world IoT network traffic dataset with 16M+ records captured over two months from diverse smart devices.	ICMP/SYN/UDP Flood, SQL Injection, Backdoor, Password Cracking, ARP, ICMP	≈1.2 GB	[368,369]
AMQP Network Traffic Dataset [167]	2025	IoT network traffic dataset over the AMQP protocol including both normal and various malicious communications.	MITM, DoS, Spoofing, Injection	≈1.23 GB	[167,370]

Table 14

Distribution of samples in ROUT-4-2023 dataset.

Attack type	Malicious	Normal	Total
Blackhole	134,282	269,852	404,134
Flooding	135,576	263,206	398,782
DODAG Version	170,242	297,818	468,060
Decreased Rank	139,891	229,108	368,999
Total	579,991	1,059,984	1,639,975

This may be due to its sensitivity to overfitting when applied to highly imbalanced datasets without sufficient noise. On the other hand, traditional models such as Logistic Regression and Gaussian Naive Bayes shows significantly lower performance, with accuracies around 72%. These models likely struggles to capture the complex and non-linear relationships among the features present in RPL attack behaviors. In particular, low recall values suggest these classifiers miss a substantial portion of the attack samples, which can be dangerous in a security context. AdaBoost performs modestly with 81.53% accuracy, demonstrating moderate effectiveness in handling the binary classification

Table 15

Extracted features from ROUT-4-2023 dataset.

Feature	Description
TIME	Simulation time
SOURCE	Source node IP
DESTINATION	Destination node IP
LENGTH	Packet length
INFO	Packet information
TR	Transmission rate (packets per 1000 ms)
RR	Reception rate (packets per 1000 ms)
TAT	Average transmission time
RAT	Average reception time
TPC	Transmitted packet count per second
RPC	Received packet count per second
TTT	Total transmission time
TRT	Total reception time
DAO	Number of DAO packets
DIS	Number of DIS packets
DIO	Number of DIO packets

problem, but again falling short of tree-based ensemble counterparts in both precision and recall.

Table 16

Performance metrics on ROUT-4-2023 dataset.

Algorithm	Accuracy (%)	Recall (%)	Precision (%)	F1 score (%)
AdaBoost	81.53	72.26	74.70	73.44
KNN	99.80	99.73	99.70	99.72
Random Forest	99.98	99.97	99.97	99.97
Decision Tree	99.99	99.99	99.99	99.99
Bagging	99.99	99.99	99.99	99.99
Logistic Regression	72.46	50.91	63.89	56.67
Gaussian NB	71.73	53.65	61.49	57.31
Gradient Boosting	90.65	90.64	84.14	87.27

Overall, these findings suggest that ensemble based tree classifiers are the most suitable choice for high-accuracy intrusion detection in IoT ecosystems. They combine interpretability, robustness, and precision, making them well-aligned with the security requirements of RPL networks.

4. Gaps in the state of the art

Despite extensive research on intrusion detection systems tailored for IoT infrastructures, significant limitations persist across both protocol layers and broader design principles [378]. At the physical layer, although threats such as jamming, eavesdropping, and signal manipulation are well documented, they remain inadequately addressed in existing IDS frameworks. These threats are often generically classified under denial-of-service categories, with limited attention to the unique signal characteristics of wireless communication. Parameters such as received signal strength indicator, signal-to-noise ratio, and channel occupancy patterns are underutilized, despite their potential to reveal physical layer anomalies. Moreover, the temporal and spectral behaviors of constrained wireless protocols such as frequency hopping and channel blacklisting are rarely exploited for detection purposes. Another critical deficiency is the lack of empirical studies considering hardware limitations, such as the energy consumption and latency introduced by continuous radio monitoring. This is particularly crucial in low power IoT deployments, where energy efficiency must be balanced with detection accuracy. Furthermore, side effects of physical layer attacks such as increased re-transmissions or MAC-layer de-synchronization are seldom correlated across layers within IDS logic. A summary of these and other major research gaps is illustrated in Fig. 9, which highlights eight underexplored or underdeveloped areas in current IDS research for IoT. As the figure illustrates, a holistic view of research gaps reveals pressing needs not only in physical layer detection but across the entire IoT contexts. Building on this, the following sections examine how such gaps manifest at higher layers.

At the data link layer, while the literature acknowledges MAC layer vulnerabilities, few IDS models provide concrete mechanisms to detect threats that exploit time synchronized and resource constrained communication patterns. Attacks such as backoff manipulation, timing based spoofing, and selective interference are often discussed only conceptually. Deterministic scheduling protocols offer opportunities for anomaly detection via deviations in slotframe usage, unauthorized timeslot access, or irregular frame transmissions, yet these indicators remain underexplored. Additionally, most systems fail to correlate MAC layer anomalies with higher-layer events, limiting the ability to detect low rate, persistent threats that evade conventional triggers. At the network layer, attacks targeting the RPL protocol such as sinkhole formation, rank manipulation, and version spoofing have received considerable attention. However, existing solutions often address these threats in isolation and rely on assumptions that are impractical in real-world deployments, such as global DODAG knowledge or ideal parent selection. There is a lack of IDS models that detect compound routing anomalies or account for protocol-specific dynamics, such as trickle timer variations and local repair events, which may obscure malicious behavior.



Fig. 9. Key research gaps in IoT intrusion detection systems.

Transport layer security remains an underexplored domain in IoT intrusion detection. While traditional transport layer attacks such as SYN flooding or session hijacking have been extensively studied in TCP based networks, these are not directly applicable to IoT protocols that primarily rely on UDP, such as CoAP or DTLS. Instead, these lightweight protocols introduce unique communication patterns including confirmable messages, stateless exchanges, and minimal handshake mechanisms that can be exploited to disrupt reliability or drain node resources. However, IDS frameworks rarely incorporate features such as repeated handshake failures in DTLS, abnormal retransmission behavior in CoAP confirmable messages, or delayed acknowledgments into their detection logic. Moreover, the limited availability of labeled datasets reflecting transport layer anomalies specific to UDP based IoT protocols further restricts progress in this area. At the application layer, intrusion detection faces the dual challenge of encrypted communication and limited semantic visibility. Protocols like CoAP, MQTT, and XMPP are widely used in constrained environments but provide minimal inspection opportunities due to end-to-end encryption (OSCORE). Consequently, IDS models are typically unable to detect payload-level attacks, such as falsified sensor readings or unauthorized commands. Although some studies propose behavioral analysis using metadata such as access frequency or timing patterns, these remain limited in scope and accuracy particularly for safety critical systems where semantic integrity is essential.

Beyond individual layers, a prominent limitation in the literature is the lack of unified, cross-layer IDS architectures. Most approaches are narrowly focused on a specific layer or protocol, missing the contextual relationships that could enhance detection accuracy. Additionally, current systems emphasize raw detection performance without adequately addressing interpretability, energy efficiency, or deployment feasibility. Deep learning based IDS often achieve high accuracy but require computational resources that are infeasible for embedded IoT devices. Furthermore, IDS models rarely incorporate explainable AI approaches to provide human-understandable rationales behind alerts. This limits operational trust and slows post-incident analysis, especially in regulated or mission-critical sectors such as healthcare or industrial IoT. While federated learning has emerged as a privacy-preserving training paradigm, studies often ignore non-IID data distributions, communication latency, and adversarial poisoning risks, which can compromise

model robustness [379–381]. The assumption of honest participants in collaborative learning scenarios is overly optimistic and presents a critical blind spot. Another issue is the widespread reuse of classical IDS datasets such as NSL-KDD and CICIDS2017, which do not reflect the traffic heterogeneity and protocol diversity of modern IoT networks. Newer datasets like CICIoT2023, Edge-IIoTset, IDSIoT2024, and X-IIoTID are rarely employed in benchmarking, and few works attempt cross-dataset evaluations to assess generalizability. There is also insufficient investigation into attack surface manipulation, where adversaries subtly alter network timing, frequency, or topology to evade anomaly detection without triggering clear violations thus introducing a false sense of security. Finally, many IDS techniques are ported from traditional IT infrastructures with minimal adaptation to IoT-specific constraints [382]. Many IDS models assume cloud-level resources, which are incompatible with the memory and energy constraints of embedded IoT devices.

Furthermore, while post-quantum resilience has primarily focused on cryptographic primitives, there is a growing body of research exploring the use of quantum computing itself as an enabler for intelligent intrusion detection. Preliminary works such as QuantumNetSec [383] and QML-IDS [384] have demonstrated that quantum machine learning techniques, including variational quantum circuits and quantum-enhanced classifiers, can significantly improve detection accuracy and training efficiency for network security tasks. Similarly, Barati [385] proposed a quantum genetic algorithm enhanced IDS tailored for wireless sensor networks, showing improved self-supervised detection under limited-label constraints. However, their adaptation to IoT environments remains extremely limited. Current QML models are typically evaluated in high-resource cloud settings or simulators, without consideration for energy-constrained, latency-sensitive edge deployment scenarios. There is thus a clear gap in understanding how quantum-enhanced ML/DL methods can be pragmatically integrated into IoT IDS pipelines, particularly in hybrid quantum-classical architectures suitable for embedded inference. In parallel, the integration of blockchain with deep learning in IDS design has gained momentum as a means of enhancing trust, traceability, and decentralization. Recent studies [386–389] propose architectures that leverage smart contracts, immutable audit trails, and federated model training to create tamper-resistant and privacy-preserving IDS frameworks [390]. Additionally, efforts like BAC-IDS [112] demonstrate the potential of blockchain assisted clustering to coordinate distributed detection logic, while others [391] employ blockchain ledgers to propagate threat intelligence across IoT domains [392]. Despite their promise, these approaches face several challenges when applied to real-world IoT systems [393, 394]. These include the computational cost of consensus mechanisms, compatibility with lightweight IoT protocols, scalability across heterogeneous device types, and the lack of standardized benchmarking practices. Moreover, few studies rigorously address the integration of blockchain-layer anomaly signals with ML based IDS decision layers in a modular and explainable fashion [395]. Beyond their role as secure logs, blockchain technologies offer foundational capabilities for building decentralized IDS architectures, especially in highly distributed IoT settings. By eliminating single points of failure and enabling trustless cooperation across nodes, blockchain can coordinate detection, logging, and response actions through consensus mechanisms such as practical Byzantine fault tolerance or proof of authority [396–398]. These mechanisms ensure that no single device controls the outcome of threat detection, thus increasing system resilience against insider attacks or compromised nodes. Moreover, smart contracts autonomous scripts that execute on-chain can be designed to trigger specific mitigation workflows (e.g., isolating a node or initiating a secure update) upon detection of predefined threat patterns. This not only accelerates response times but also ensures consistent policy enforcement across heterogeneous devices. Blockchain's immutability further supports forensic traceability, while integration with privacy-enhancing techniques such as ECC, DSA, enables secure data exchange without

compromising confidentiality. Additionally, systems such as InterPlanetary file system can be leveraged alongside blockchain to provide distributed storage for large-scale IDS telemetry, enabling scalable and tamper-proof data sharing among federated or geographically separated IoT deployments [223]. These architectural capabilities remain underutilized in current IDS research, which largely treats blockchain as a passive logging layer rather than an active participant in decision-making. As such, there is a pressing need to explore blockchain based IDS frameworks that tightly integrate consensus, automation, and secure data flows, while maintaining compatibility with constrained IoT protocols. As a result, blockchain based AI approaches in intrusion detection remain promising but largely fragmented, requiring further research into unified architectural patterns and runtime efficiency in resource constrained environments [113,399].

In conclusion, while substantial progress has been made in developing IDS solutions for IoT environments, critical gaps remain in layer specific coverage, cross layer correlation, real-world applicability, and methodological transparency. Future research should prioritize protocol-aware, lightweight, explainable, and resilient IDS models that are validated using diverse, modern datasets and that operate effectively within the realistic limitations of edge and embedded IoT contexts. Moreover, long-lifespan IoT systems must account for future cryptographic disruptions, especially those posed by emerging quantum threats. While most IDS frameworks target current classical threats, they remain largely unprepared for post-quantum security scenarios [54]. Quantum computing poses a significant future risk to conventional cryptographic primitives such as RSA, ECC, and even certain symmetric key algorithms at reduced key lengths [400]. However, lightweight IDS models especially those deployed on constrained devices rarely incorporate quantum-resistant cryptographic mechanisms or anticipate quantum-resilient attack surfaces [401,402]. Given the long-term deployment lifecycle of many IoT systems, ignoring post-quantum readiness represents a strategic oversight. Integrating post-quantum cryptographic standards or evaluating the impact of Shor's and Grover's algorithms on IDS decision layers remains an open research direction.

5. Challenges and future directions

Despite significant progress in IoT intrusion detection systems, several key challenges remain unsolved, particularly when transitioning from experimental setups to real-world deployments. Many existing IDSs perform well under controlled conditions but fail to maintain effectiveness in heterogeneous, resource-constrained, and dynamic IoT environments. One persistent challenge is the ability to detect anomalies using incomplete, noisy, or encrypted data without compromising system performance or user privacy. As described in Section 3.3, anomaly based IDS approaches rely heavily on behavioral profiles and statistical baselines. These methods are vulnerable to false alarms and degraded accuracy when data quality is poor a common condition in IoT due to limited device instrumentation and intermittent connectivity. To address this, future IDS should leverage richer data sources, including device-level provenance, control signal traces, and system context extracted through lightweight monitoring. This approach is especially critical in time-sensitive applications like industrial control systems or smart healthcare, where anomaly attribution must be both fast and precise.

Adaptability is another essential requirement, but it must go beyond incremental learning. As shown in hybrid approaches discussed in Sections 3.5.1, and 3.5.2, combining rule based detection with anomaly based learning allows the system to handle known threats while adjusting to evolving patterns. However, many of the existing frameworks treat adaptation as periodic retraining, often ignoring runtime feedback or domain semantics. These mechanisms should be natively designed for edge deployment, considering practical constraints such as CPU cycles, memory footprint, and energy consumption. The role of AI in

IDS design is growing, yet its practical deployment remains limited. As outlined in Section 3.3, advanced models such as GANs, LSTMs, and federated learning architectures show promise in offline evaluations. However, their runtime complexity and hardware demands restrict their use in embedded environments. Instead of porting cloud-based models to the edge, future work should focus on designing energy-efficient, hardware-aware AI tailored for low-power IoT devices.

Another often-overlooked challenge is the data collection layer. As shown in Section 2.1, effective IDS design depends heavily on the quality and granularity of data gathered across the physical, network, and application layers. However, diverse device architectures and inconsistent data formats make standardization and integration difficult. Continuous logging drains device resources, while sparse logging reduces detection fidelity. These constraints are also reflected in the publicly available datasets surveyed in Section 3.6, where many datasets lack real-world diversity, contain outdated traffic patterns, or do not represent all IoT layers comprehensively. This limits the generalizability and robustness of IDS models trained on them. Future work should prioritize event-driven, selective data sampling techniques and edge-side preprocessing to reduce transmission overhead while maintaining semantic richness. In addition to challenges rooted in data representation and system resource constraints, emerging technological shifts such as quantum computing introduce new risks that must be proactively addressed. As quantum technology continues to evolve, it poses a serious threat to widely used cryptographic algorithms that secure IDS communications and device authentication processes. Preparing IDS frameworks for a post-quantum future will require more than simply adopting quantum-resistant encryption. It will also necessitate rethinking trust models, secure software update mechanisms, and long-term key management strategies especially in distributed IoT environments where devices may operate unattended for years. Anticipating these risks early can help ensure long-term resilience. Complementing these risks, emerging paradigms such as quantum-enhanced machine learning and blockchain based trust architectures offer promising but underexplored directions for next-generation IoT IDS. While prior work has demonstrated the theoretical potential of QML based classifiers and federated blockchain-integrated frameworks, future IDS development must focus on translating these into deployable, resource-aware solutions. For QML, this includes designing hybrid inference pipelines that balance quantum circuit complexity with edge latency constraints, and assessing resilience under low-label or adversarial settings. On the blockchain side, practical deployments will require rethinking consensus overhead, integrating distributed ledger telemetry with anomaly detectors, and ensuring cross-layer explainability. Both directions also demand new benchmarking protocols and toolchains tailored to quantum or decentralized architectures, as existing IDS evaluations do not capture their unique system-level behaviors. In particular, blockchain offers more than just tamper-proof logging and decentralized model training. Its integration into IDS architectures enables secure, autonomous, and distributed decision-making across heterogeneous IoT environments. Smart contracts in blockchain can be used to trigger predefined actions such as node isolation, firmware rollback, or distributed alert propagation, based on consensus-driven threat validation. This reduces dependence on centralized controllers and enhances system responsiveness to emerging threats. Furthermore, the immutable ledger maintained by blockchain nodes can serve as a trusted audit trail for forensic analysis and compliance enforcement, especially in critical sectors such as healthcare and smart manufacturing. Cryptographic primitives such as ECC and digital signatures ensure that IDS related transactions (threat alerts, model updates) are authenticated and traceable. Additionally, privacy-preserving data sharing can be achieved using blockchain in conjunction with off-chain storage frameworks. This is particularly relevant in federated or cross-domain IDS scenarios, where sensor level data needs to be shared securely without violating regulatory constraints. Despite these advantages, real-world adoption remains limited due to latency introduced by consensus

protocols, energy overhead of blockchain maintenance, and integration challenges with lightweight IoT stacks. Therefore, future IDS research should explore hybrid blockchain architectures that decouple control flow from heavy consensus operations, leverage lightweight consensus, and support modular interoperability with existing IoT environment.

In summary, building the next generation of IDS for IoT requires a shift from narrow performance optimization to resilient, context-aware, and system-integrated designs. These systems must operate within real-world limitations while leveraging advances in AI, edge computing, and cryptography. Most importantly, progress will depend on sustained collaboration across disciplines including embedded systems, cybersecurity, and network engineering to ensure that proposed solutions are not only technically sound, but also scalable, trustworthy, and ethically grounded.

6. Conclusion

This study presented a comprehensive analysis of IDS tailored for the IoT, examining the core detection paradigms including signature based, anomaly based, specification based, and hybrid approaches. Through this review, the strengths, limitations, and application contexts of each detection methodology are outlined, highlighting their roles in enhancing IoT network security. The findings emphasize the pivotal contribution of machine learning and deep learning techniques in advancing IDS performance, particularly in improving detection accuracy and adaptability to dynamic threat landscapes. Nevertheless, challenges such as resource constraints, heterogeneous device environments, and the evolving sophistication of cyber-attacks necessitate continuous innovation in IDS design. Our review also highlights the growing importance of cross-layer detection strategies, explainable AI, and the integration of post-quantum cryptographic mechanisms to future-proof IoT security systems. To effectively safeguard future IoT ecosystems, ongoing efforts must prioritize the development of lightweight, adaptive, and privacy preserving IDS architectures. Furthermore, interdisciplinary collaboration across security, networking, and artificial intelligence domains will be crucial in addressing the complex security requirements of next-generation IoT applications. In conclusion, while significant progress has been achieved in IDS research for IoT, the dynamic nature of security threats demands continuous evolution. Next-generation IDS must not only address current vulnerabilities but also anticipate emerging risks such as quantum-enabled cryptanalysis and ensure robustness through validation with modern, representative datasets. Striking the right balance between detection efficacy, computational efficiency, and scalability will be essential to secure the expanding IoT landscape.

CRedit authorship contribution statement

Burak Aydin: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Hakan Aydin:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Visualization. **Sedat Gormus:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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